AVERAGE:→

① The average of x numbers is y and average of y numbers is x. The average of y numbers is x. The average no. taken together is

① $\frac{x+y}{2xy}$ ② $\frac{2xy}{x+y}$ ③ $\frac{x^2+y^2}{x+y}$ ④ $\frac{xy}{x+y}$

Sum of no = xy Average = $\frac{xy+yx}{x+y}$ $\frac{2xy}{x+y}$

" " " = yx

② A library has an average no. of 510 visitors on Sunday and 240 on other days. The avg no of visitors per day in a month of 30 days beginning with Sunday is.

① 285 ② 295 ③ 300 ④ 290

Sundays in a month = 5 ~~25×51~~ $\frac{25\times240 + 510\times5}{30} = \frac{8550}{30}$

= 285

③ The avg of 8 no is 27. If each of the no is multiplied by 8 find the avg of new set of ~~num~~ no.

① 1128 ② 938 ③ 316 ④ 216

If each no. is multiplied by 8 then avg will also become multiple of

$27\times8 = 216$

④ There are 100 students in 3 sections A, B, C of a class. The avg marks of all 3 ~~no~~ is sections is 84. The avg of B and C = 87.5 and avg marks of A = 70 The no. of students in class A

① 30 ② 35 ③ 20 ④ 25

~~A+B~~ B+C 87.5 A 70

84

14 3.5

4 : 1

5 units = 100

1 unit = 20

⑤ The avg of some natural no. is 15. If 30 is added to 1st no and 5 sub from the last no. the avg becomes 17.5 then the no. of natural number is

① 15 ② 30 ③ 20 ④ 10

$15\times x + 30-5 = 17.5\times x$

~~2.5x~~ = 25 = 2.5x

(6) The avg weight of 15 oarsmen in a boat is increased by 1.6 kg when one of the crew, who weighs 42 kg is replaced by a new man. Find the weight of the new man.

(1) 67 (2) 65 (3) 66 (4) 43

weight of new man = $(42 + 15 \times 1.6) = 66$ kg

R = 42 M = 15 I = 1.6 (R + M × I)

(7) Three science classes A, B, C taken a life science test. The avg score of class A = 83 B = 76. The avg score C = 85 A+B = 79 B+C = 81 Avg of A, B, C = ?

(1) 81.5 (2) 80.5 (3) 81 (4) 80

83, 76 → 79 → 3, 4 76, 85 → 81 → 4, 5

A : B : C = 3 : 4 : 5

Avg of A, B, C = $\frac{83 \times 3 + 76 \times 4 + 85 \times 5}{3+4+5} = 81.5$

(8) The Avg marks obtained by 100 candidates in a examination is 30. If the avg of passed = 35 and failed = 10. The no. of candidates who passed in the examination?

(1) 60 (2) 70 (3) 80 (4) 20

35, 10 → 30 → 20, 5

passed = $\frac{4}{5} \times 100 = 80$

of passed = x failed = 100 − x

$35 \times x + (100 - x) \times 10 = 3000$

$35x + 1000 - 10x = 3000$

$25x = 2000$

$\boxed{x = 80}$

(9)

(25) The avg marks of 17 students was calculated as 71. But it was later found that mark of one student had been wrongly entered as 65 instead of 56 and another 24 instead of 50. The correct avg is

(1) 70 (2) 71 (3) 72 (4) 73

correctly entried = 56 + 50 = 106 106 − 89 = 17

wrongly entered = 65 + 24 = 89 $\frac{71 + 17}{17}$ = 71 + 1 = 72

(10) A group of boys has an avg weight of 36 kg. one boy weighting 42 kg leaves the group and another boy weighting 30 kg joined the group. If the avg now becomes 35.7 kg then how many boys are there in the group

(1) 30 (2) 32 (3) 40 (4) 56

$36x = 35.7x + (42 - 30)$ $36x + 42 - 30 = 35.7x$

$0.3x = 12$ $0.3x = 12$ $\boxed{x = 40}$

$\boxed{x = 40}$

(11) What is the avg of all no. between 100 and 200 which are divisible by 13

(1) 147.5 (2) 145.5 (3) 143.5 (4) 149.5

Smallest no divisible by 13 = 104

Largest no divisible by 13 = 195

Avg = $\frac{a + l}{2} = \frac{104 + 195}{2} = \frac{299}{2} = 149.5$

(12) The avg weights of L, M, N are 93 kg. The avg L+M = 89 M+N = 96.5 then weights in kg of M is

(1) 92 (2) 86 (3) 101 (4) 95

L+M = 188   L+M+N = 279

M+N = 193   188 + N = 279   M+N = 193

N = 101 $\boxed{M = 92}$

(13) The avg of 2 no is 10 and the product of 2 no is 96. Find the 2 numbers.

(1) 3, 32 (2) 12, 8 (3) 1, 96 (4) 24, 4

$\frac{x+y}{2} = 10 \Rightarrow x + y = 20$ $x \times y = 96$

$\therefore (x-y)^2 = (x+y)^2 - 4xy$

$= \sqrt{(20)^2 - 4 \times 96} = x - y = \sqrt{16} = 4$

x + y = 20

x − y = 4

2x = 16 $\boxed{x = 8}$

$\boxed{y = 12}$

(14) The avg of 11 no is 54. The avg of 1st 4 no is 48, and that of that the next 4 no is 25% more than the average of the first four. The ninth no is 8 greater than the 11th no and the 10th no is 4 greater than 11th no. What is the average of the 9th and 10th numbers?

① 54 ② 52.6 ③ 56 ④ 54.4

Let 11th no is = x

9th no is = x+8

10th no = x+4

$54 \times 11 = 594$

$\frac{48}{192} \times \frac{1}{4} = 48$; $192 + 48 = 240 \rightarrow$ next 4

$240 + 192 = 432$

Total = $240+192+x+x+8+x+4 = 594$

$432 + 3x + 12 = 594$

$444 + 3x = 594 \Rightarrow 3x = 150$

$x = 50$

$594 - 444 = 150$

Avg of 9th and 10th = $\frac{58+54}{2} = 56$

(15) The avg of 33 no is 74. The avg of 1st 17 no is 72.8 and that of last 17 no is 77.2. If the 17th no is excluded, then what will be the avg of remaining no.

① 72.9 ② 73.4 ③ 71.6 ④ 70.8

$17 \times 72.8 + 17 \times 77.2 - 33 \times 74$

$= 17(72.8 + 77.2) - 2442$

$= 17 \times 150 - 2442 = 108$

Sum of the no excluding 108 = $2442 - 108 = \frac{2334}{32} = 72.9$

(16) The avg of n no is 36. If each of 75% of the no. is increased by 6 and each of the remaining no is decreased by 9, then new average of the no is

① 37.125 ② 33.75 ③ 39.25 ④ 36.25

$= \frac{\frac{3n \times 6}{4} - \frac{n \times 9}{4}}{n} = \frac{\frac{18n}{4} - \frac{9n}{4}}{n} = \frac{9}{4} = 2.25$

New avg = $36 + 2.25 = 38.25$

TYPE-2 : Problems on consecutive odd and even no.'s

(17) The avg of 5 consecutive integers starting with 'm' is n. What is the avg of 6 consecutive integers starting with (m+2)?

① $\frac{2n+5}{2}$ ② (n+2) ③ (n+3) ④ $\frac{2n+1}{2}$

① $m + m+1 + m+2 + m+3 + m+4$

$\frac{5m+10}{5} = n$

$5m + 10 = 5n$

$m + 2 = n$

$m+2 + m+3 + m+4 + m+5 + m+6 + m+7$

$\frac{6m+27}{6} = 6$

$= \frac{2m+9}{2} = \frac{2(n-2)+9}{2} = \frac{2n-4+9}{2}$

$= \frac{2n+5}{2}$

(18) The avg of 25 consecutive odd integers is 55. The highest of these integers is.

① 79 ② 105 ③ 155 ④ 109

The avg of 25 consecutive odd integers = 55

Mid no. i.e 13th = 55, so 25th no = $55 + 2 \times 12$ $= 79$

(19) The avg of 39 no is 2000. Out of them, how many may be greater than 2000, at the most?

① 20 ② 0 ③ 38 ④ 39

The avg of 39 no = 0

Out of them 38 may be greater than 2000 at the most

(20) The avg of first 1234 ___ no's is equal to 1234 Fill the blank

① odd ② even ③ prime ④ Natural

The avg of 1st 2 odd no = $\frac{1+3}{2} = \frac{4}{2} = 2$

1st 3 " " = $\frac{1+3+5}{3} = \frac{9}{3} = 3$

1st 4 odd no = $\frac{1+3+5+7}{4} = \frac{16}{4} = 4$

hence 1234 odd no's is equal to 1234

(21) Fill in the blank

The avg of 1st 101 ___ numbers is equal to 102.

① Natural ② odd ③ even ④ perfect square

1st 2 even no = $\frac{2+4}{2} = \frac{6}{2} = 3$ (2+1)

3 even no = $\frac{2+4+6}{3} = \frac{12}{3} = 4$ (3+1)

hence the 1st 101 no is equal to (101+1) = $\underline{102}$

(22) The difference b/w avg of 1st ten prime no's and 1st ten prime no of 2 digits is

① 14.5 ② 16.5 ③ 12.5 ④ 13.5

$\frac{2+3+5+7+11+13+17+19+23+29}{10} = \frac{129}{10} = 12.9$

$\frac{11+13+17+19+23+29+31+37+41+43}{10} = \frac{264}{10} = 26.4$

$26.4 - 12.9 = \cancel{26.5}\ 13.5$

(23) Type:3 find nth no when avg of 1st no p and last q no are given

(23) The avg of largest and smallest 3 digit no's formed by 0, 2 and 4 would be

① 312 ② 212 ③ 222 ④ 303

Largest = 420 but smallest ~~024~~ 204

$\frac{420+204}{2} = \frac{624}{2} = \underline{312}$

(24) Of the three no, the first no is twice of the second and the second no is 2 thrice of the 3rd number. If the average of 3 no's is 20. then the sum of the largest and smallest no is

① 24 ② 42 ③ 54 ④ 60

$\frac{x+y+z}{3} = 20 = x+y+z = 60$

$x = 2y$ $y = 3z$

$x+y+z = 60$

$2y + y + \frac{y}{3} = 60$

$6y + 3y + y = 180$

$10y = 180$

$\boxed{y = 18}$

$x + y = \underline{42}$

$\Rightarrow x = 36$

$y = 6$

TYPE-4 Imp To find correct avg of no's earlier some mistakes has made while entering

(25) The avg of 25 observations is 13. It was later found that an observation 73 was wrongly entered as 48. The new avg is

① 12.6 ② 14 ③ 15 ④ 13.8

avg 25 = 625×13 ... = 325

Difference = 73 − 48 = 25

New avg = $13 + \frac{25}{25} = \underline{14}$

or

old = 325 New = 325 + 73 − 48 = 325 + 25 = 350

$\frac{350}{25} = \underline{14}$

(26) The mean of 100 items was 46. Later on it was discovered that an item 16 was misread as 61 and another item 43 was misread as 34. It also found that the number of items were 90 and not 100. Then what is the correct mean

① 50 ② 50.7 ③ 52 ④ 52.7

100×46 = 4600

34 − 61 = −27 43 − 34 = 9

= 27 + 9 = 36

$\frac{4600-36}{90} = \frac{4564}{90} = \underline{50.7}$

(27) The avg marks obtained by 40 students of class is 86. If the 5 highest marks were removed, the average reduces by one mark. The avg marks of the top 5 students is

① 92 ② 96 ③ 93 ④ 97

86×40 = 3440

3440 − 5x = 85×35

3440 − 5x = 2975

465 = 5x

$\boxed{x = 93}$

(28) While tabulation of marks scored in an examination by the students of a class, by mistake the marks scored by one student got recorded as 93 in place of

63, and thereby the average marks increased by 0.5. What was the number of students in a class

① 20 ② 20 ③ 15 ④ 60

(i) correct score = 63 (ii) Incorrect score = 93

$93-63=30$

Number of students $= \frac{30}{0.5} = \underline{60}$

Type: 6

(29)

① In a class, the average score of girls in an examination is 73 and that of boys is 71. The average score for whole class is 71.8 Find the percentage of girls.

① 40% ② 50% ③ 55% ④ 60%

73 71

71.8

0.8 1.2

2 : 3

No. of girls % $= \frac{2}{5}\times 100 = 40\%$

(30) If the mean of 4 observations is 20, when a constant 'c' is added to each observation, the mean becomes 22. The value of c is

① 6 ② -2 ③ 2 ④ 4

$4\times20=80$

$80+c=22\times5$

$80+c=110$

$c=22-20$

$4c=22\times4-20\times4$

$4c=88-80$

$\boxed{c=2}$

(31)

Type 7 → Cricket

(31) The batting average of a cricket player for 64 innings is 62 runs. His highest score exceeds his lowest score by 180 runs. Excluding these two innings, the average of remaining innings becomes 60 runs. His highest score was

① 180 runs ② 209 runs ③ 212 runs ④ 214 runs

64×62

128

3844

3968

Highest $=x$

Lowest $=y$

$x=y+180$

$3968-60\times62$

$3968-4960$ 3720

4960

3968

992

3968

3720

248

$x+y=248$

$y+180+y=248$

$2y=428$

$y=214$

248

180

428

(32) The batting average for 30 innings of a cricket player is 40 runs

(32) Sachin Tendulkar has a certain average for 11 innings. In the 12th innings he scores 120 runs and thereby increases his average by 5 runs. His new average is.

① 60 ② 62 ③ 65 ④ 66

$11x+120=12(x+5)$ New average $=x$

Total runs in 11 innings $=11(x-5)$

$11(x-5)+120=12x$

$11x-55+120=12x$

$\boxed{x=65}$

(33) A batsman makes a score of 81 runs in the 16th match and thus increases his average runs per match by 3. What his average after 16th match?

(1) 35 ② 33 ③ 34 ④ 36

$15(x-3)+81=16x$

$15x-45+81=16x$

$\boxed{x=36}$

Type: VIII Problems of ages

(34) The average age of a family of 10 members is 20 years. If the age of the youngest member

of a family is 10 years, then the average age of the members of the family just before the birth of the youngest member was approximately

① 27.14 years ② 12.5 years ③ 14.28 yrs ④ $11\frac{1}{9}$ yrs

present age = $10 \times 20 = 200$ yrs
10 yrs before = $10 \times 10 = 100$
Required Average = $\frac{100}{9} = 11\frac{1}{9}$

(35) The avg age of husband and his wife was 23 years at the beginning of their marriage. After 5 years they have a one-year old child. The average age of the family of three. when the child was born?

① 23 yrs ② 24 yrs ③ 17 yrs ④ 20 yrs

$\frac{H+W}{2} = 23$
$H+W = 46$ after 5 years $H+W = 56 + \text{child} = 57$
$57 - 3 = \frac{54}{3} = \underline{18}$

(36) The avg age of Ram and his two children is 17 yrs and the avg of Ram's wife and same children is 16 yrs If the age of Ram is 33 years. the age of his wife is

① 31 ② 32 ③ 35 ④ 30

$\frac{R+2c}{3} = 17 = R+2c = 51$
$W+2c = 48$
$R-W = 3$
$33 - W = 3$
$\boxed{W = 30}$

(37) In the family of 5 members, the average age at present is 33 years The youngest member is 9 yrs old. The avg age of the family just before the birth of the youngest member was

① 30 yrs ② 29 yrs ③ 25 yrs ④ 24 yrs

$33 \times 5 = 165$
9 yrs old = $9 \times 5 = 45$
$165 - 45 = 120 \rightarrow$ 9 yrs back age
$\frac{120}{4} = \underline{30}$

(38) 3 years ago, the average age of a family of 5 members was 17 yrs. A baby having been born. The avg age of the family is same today. The present age of the baby is

① 1 years ② $1\frac{1}{2}$ years ③ 2 years ④ 3 years

$17 \times 5 = 85$ · $5 \times 3 = 15$
present = $85 + 15 = 100$

$100 + x = 17 \times 6$
$100 + x = 102$
$x = 2$ years

(39) The avg age of a cricket team of 11 players is the same as it was 3 years back because 3 of the players whose current avg was 33 yrs were replaced by 3 youngsters. The avg age of the newcomers is

① 23 years ② 21 years ③ 22 years ④ 20 years

$33 \times 3 - 11 \times \frac{3}{3}$
$99 - 33 = \frac{66}{3} = 22$ years

(40)

PERCENTAGE :→

Concept :→

35% of 80 marks $= \frac{35}{100} \times 80 = 28$

Percentage to Fraction form

$\frac{35}{100} = \frac{7}{20}$ from $5\% = \frac{5}{100} = \frac{1}{20}$ $\xrightarrow{\times 7}$ $\frac{7}{20} = 35\%$

$\frac{5}{100} = \frac{1}{20}$ $65\% = \frac{13 \times 5}{20} = \frac{13}{20}$

Multiples of 5 $85\% = \frac{17 \times 5}{20} = \frac{17}{20}$

Percentage to fraction values

$\frac{1}{2} \longrightarrow 50\%$

$\frac{1}{3} \longrightarrow 33.33\% = 33\frac{1}{3}\%$

$\frac{1}{4} \longrightarrow 25\%$

$\frac{1}{5} \longrightarrow 20\%$

$\frac{1}{6} \longrightarrow 16.66\% = 16\frac{2}{3}\%$

$\frac{1}{7} \longrightarrow 14.28\% = 14\frac{2}{7}\%$

$\frac{1}{8} \longrightarrow 12.5\% = 12\frac{1}{2}\%$

$\frac{1}{9} \longrightarrow 11.11\% = 11\frac{1}{9}\%$

$\frac{1}{10} \longrightarrow 10\%$

$\frac{1}{11} \longrightarrow 9.09\% = 9\frac{1}{11}\%$

$\frac{1}{12} \longrightarrow 8.33\% = 8\frac{1}{3}\%$

$\frac{1}{13} \longrightarrow 7.69\% = 7\frac{9}{13}\%$

$\frac{1}{14} \longrightarrow 7.14\% = 7\frac{1}{7}\%$

$\frac{1}{15} \longrightarrow 6.66\% = 6\frac{2}{3}\%$

$\frac{1}{16} \longrightarrow 6.25\% = 6\frac{1}{4}\%$

$\frac{1}{20} \longrightarrow 5\%$

$\frac{1}{25} \longrightarrow 4\%$

$\frac{1}{20} - \frac{1}{50} \to 2\%$

Ex:- ~~$54\frac{6}{11}$~~ $54\frac{6}{11}\%$ of 132

$\frac{1}{11} = 9\frac{1}{11} \xrightarrow{6} 54\frac{6}{11} \times 132 =$ $\left[9\frac{1}{11} = \left(9+\frac{1}{11}\right)\%\right]$

Ex:2 $23\frac{1}{13}\%$ of $78 = ?$

$\frac{1}{13} \to 7.69 = 7\frac{9}{13}$

$= \left(7 + \frac{9}{13}\right) \times 3 = \left(21 + \frac{27}{13}\right)$ $13)27(2$ $\frac{26}{1}$ $= 2\frac{1}{13}$

improper fraction

$\left(21 + \frac{27}{13}\right)$ $\left(2\frac{1}{13}\right)$

$23\frac{1}{3}$ as multiples of 3 of $\frac{1}{13}$

$\frac{3}{13} \times 78 =$ Ans

Ex:-3 $54\frac{1}{7}\%$ of 147

$\frac{1}{7} = \left(14\frac{2}{7}\right) \times 4 = \frac{4}{7} \times 147 =$

$\frac{1}{7} = \left(14 + \frac{2}{7}\right) = 56 + \frac{8}{7}$ $1\frac{1}{7}$

$57\frac{1}{7}$

Ex:-4 $44\frac{4}{9}\%$

$= \frac{1}{9} = \left(11\frac{1}{9}\right)^{\times 4} = 44\frac{4}{9}$

percentage comparison (% more, % less, % of)

× A is what % of B = $\boxed{\frac{A}{B} \times 100}$
(32) (80)
= $\frac{32}{80} \times 100$ =

A is what % less than B =

B is what % more than A =

$\boxed{\%\text{error} = \frac{\text{diff}}{\text{correct}} \times 100}$

A is what % of B = $\boxed{\frac{A}{B} \times 100}$
(32) (80)

= $\frac{32}{80} \times 100 = 40\%$

A is what % less than B = $\boxed{\frac{\text{diff}}{\text{more}} \times 100}$
(32) (80)

= $\frac{48}{80} \times 100 = 60\%$

B is what % more than A = $\boxed{\frac{\text{diff}}{\text{less}} \times 100}$

= $\frac{48}{32} \times 100 = 150\%$

Ex:→ 40 is what % less than 90. (in fraction form)

$\frac{\text{diff}}{\text{more}} \times 100 = \frac{5}{9} \times 100$

$= \frac{1}{9} = (11\frac{1}{9}) \times 5 = 55\frac{5}{9}\%$

Question based on percentage - Fraction & comparison

① If the income of A is 25% more than that of B. How much % B's income is less then that of A's

(a) 20% (b) 30% (c) 15% (d) 40%

soln :→

A	B
125 :	100
5	4

(or) $25\% = \frac{1}{4} \rightarrow B$ $4+1=5 \rightarrow A's$

$\%\text{less} = \frac{\text{diff}}{\text{more}} \times 100 = \frac{1}{5} \times 100 = 20\%$

② If the income of A is 40% less than that of B. How much % B's income is more then that of A's

(a) 60% (b) 40% (c) 50% (d) 66.66%

soln $40\% = \frac{2}{5} \rightarrow B's$ $5-2=3$ A's

$\text{more}\% = \frac{\text{diff}}{\text{less}} \times 100$

$= \frac{2}{3} \times 100$

$= 66.66\%$

③ Two no are less than third no. by 30% and 37% respectively. The percent by which the second no. is less than the 1st is

(a) 10% (b) 7% (c) 4% (d) 3%

soln

I	II	III
70	63	100

30% −37

$\%\text{less} = \frac{\text{diff}}{\text{more}} \times 100 = \frac{7}{70} \times 100 = 10\%$

④ If first no. is 8.33% less than third no. and ratio of second and third no is 15:16 then average of first and third no is how much % more than second no.

(a) 2.45% (b) 2.22% (c) 3.33% (d) 2.75%

8.33%

$8.33\% = \frac{25}{3} = 2\frac{1}{12}$

A	B	C
11		12) ×4
	15	16) ×3
44	45	48

LCM = 48

$\frac{92}{2} = 46$ % more $= \frac{1}{45} \times 100$

$= \frac{1}{5} \times \frac{1}{9}$

$= 20\% \times \frac{1}{9}$

$= 2.22\%$

⑤ If a student multiplied a no. by $\frac{7}{12}$ instead of $\frac{11}{16}$ find the % error in calculation.

(a) 11.11% (b) 13.33% (c) 15.15% (d) 16.16%

Correct → $\square \times \frac{11}{16}$ LCM of 16, 12 = 48 $48 \times \frac{11}{16} = 33$

Incorrect → $\square \times \frac{7}{12}$ $48 \times \frac{7}{12} = 28$

% less $= \frac{5}{33} \times 100 = \frac{5}{11} \times \frac{1}{3} \times 100$

$= \frac{5}{11} \times 33\frac{1}{3}\%$

$= 15.15\%$

⑥ Ratio between the expenditure and saving of a man is 5:4. If the income is increased by 20% and saving is decreased by 10% then calculate the % change in expenditure.

(a) 46% (b) 44% (c) 30% (d) 36%

Exp	+	Sav	=	Income	
5	:	4	:	9	×10
50	:	40	:	90	
				↓ 20% = 18	
72	+	36	=	108	

% more $= \frac{22}{50} \times 100 = 44\%$

Successive (Consit, effective, net) percentage change.

worker → salary → +40% −20%

$\boxed{100} \xrightarrow{+40\%} \boxed{140} \xrightarrow{-20\%} \boxed{112}$

+12%

Formula → $\pm x \pm y \pm \frac{xy}{100}$ + → Increase, − → Decrease

$+40 - 20 - \frac{40 \times 20}{100}$

$20 - 8 = 12\%$ — Increased by 12%

Advance :— $40\% = \frac{2}{5}$ $20\% = \frac{1}{5}$

5 : 7

5 : 4

25 : 28 → $\frac{3}{25} \times 100$

$= \frac{3}{25} \times 4$

$= 12\%$

⑦ The length and breadth of a rectangle are increased by 20% and 30% respectively by how much % the area of rectangle increased?

(a) 42% (b) 56% (c) 55% (d) 58%

$+20 + 30 + \frac{600}{100} = 50 + 6 = +56\%$ increased

⑧ If the length of the rectangle is increased by $37\frac{1}{2}\%$ and if breadth is decreased by 20%. Find the % change in area.

(a) 12.5% (b) 20% (c) 10% (d) 8.33%

$+37\frac{1}{2}\% = \frac{3}{8}$ $-20\% = \frac{1}{5}$

= 8 : 11

5 : 4

40 44

% more $= \frac{4}{40} \times 100 = 10\%$

(9) The price of sugar is increased by $16\frac{2}{3}\%$ and the consumption of a family is decreased by 25%. find % change in expenditure?

(a) $6\frac{2}{3}\%$ (b) $16\frac{2}{3}\%$ (c) 10% (d) $12\frac{1}{2}\%$

$+16\frac{2}{3} = \frac{1}{6}$ $\quad -25\% = \frac{1}{4}$ $\quad = \begin{matrix} 6 & : & 7 \\ 4 & & 3 \\ 24 & & 21 \end{matrix}$

$\%\text{ less} = \frac{3}{24} \times 100$

$= 12\frac{1}{2}\%$

(10) The sale of cinema ticket is increased by $57\frac{1}{7}\%$ and the price of ticket is increased by $16\frac{2}{3}\%$ find the change in revenue?

(a) 75% (b) $42\frac{6}{7}\%$ (c) $83\frac{1}{3}\%$ (d) 90%

$+57\frac{1}{7}\% = \frac{4}{7}$ $\quad +16\frac{2}{3}\% = \frac{1}{6}$ $\quad = \begin{matrix} 7 & : & 11 \\ 6 & : & 7 \\ 42 & & 77 \end{matrix}$

$\%\text{ more} = \frac{35}{42} \times 100$

$= \frac{1}{6} \times \frac{5 \times 100}{3} = 16\frac{2}{3}\% \times 5 = 80 + \frac{10}{3}$

$= (80)(3\frac{1}{3})$

$= 83\frac{1}{3}\%$

(15) Percentage Increase and Decrease

$+20\% \longrightarrow 120\% \longrightarrow \frac{120}{100}$ $\quad$ Ex:- $4800 \rightarrow$ 1st yr $20\% \rightarrow$ 2nd yr 25%

$+38\% \longrightarrow 138\% \longrightarrow \frac{138}{100}$ $\quad = 4800 \times \frac{120}{100} \times \frac{125}{100}$

$-40\% \longrightarrow 59\% \longrightarrow \frac{58}{100}$ $\quad = 4800 \times \frac{6}{8} \times \frac{5}{4}$

$-23\% \longrightarrow 77\% \longrightarrow \frac{77}{100}$ $\quad = 7200$

(11) The population of town increases by $6\frac{2}{3}\%$ every year, if its present population is 33750, then after 2 yrs how much will it be?

(a) 38900 (b) 38400 (c) 52090 (d) 54900

$6\frac{2}{3}\% = \frac{1}{15} \longrightarrow \frac{16}{15}$

$33750 \times \frac{16}{15} \times \frac{16}{15} = 150 \times 256 = \cancel{150 \times 10 + 15}$

$256 \times 10 + 256 \times 5$

$= 2560 + 1280$

$= 38400$

(12) If $8\frac{1}{3}\%$ of a no. is added to itself then result becomes 520. find the original no.?

(a) 480 (b) 500 (c) 420 (d) 410

$8\frac{1}{3}\% = \frac{1}{12} \xrightarrow{+} \frac{13}{12}$

$x \times \frac{13}{12} = 520$

$\boxed{x = 480}$

(13) If $6\frac{2}{3}\%$ of no. is subtracted from itself then result becomes 4200. find original no.

(a) 4500 (b) 4500 (c) 4350 (d) 4650

$6\frac{2}{3}\% = \frac{1}{15} \xrightarrow{-} \frac{14}{15}$

$x \times \frac{14}{15} = 4200 \quad = \quad x = 4500$

(14) The population of a town is increased by $16\frac{2}{3}\%$ on first year and on second year it is again increases by $37\frac{1}{2}\%$ but on third year it will decrease by $57\frac{1}{7}\%$. The find present population if after three years the population will become 1,65,000.

(a) 1,85,000 (b) 2,40,000 (c) 2,25,000 (d) 2,45,000

$+16\frac{2}{3}\% \longrightarrow \frac{1}{6} \to \frac{7}{6} + 37\frac{1}{2}\% = \frac{3}{8} \to \frac{11}{6} + 57\frac{1}{7} \to \frac{4}{7} = \frac{3}{7}$

Exam Approach

$$\boxed{x \times \frac{7}{6} \times \frac{11}{8} \times \frac{3}{7} = 1{,}65{,}000}$$

$$\boxed{x = 2{,}40{,}000}$$

(15) If the numerator of a fraction is increased by 20% and denominator is decreased by 10% the value of the new fraction become $\frac{3}{8}$. The original fraction is

(a) $\frac{24}{19}$ (b) $\frac{3}{13}$ (c) $\frac{9}{32}$ (d) $\frac{7}{9}$

fraction $= \frac{x}{y}$ $\quad 20\% = \frac{1}{5} \to \frac{6}{5} \quad 10\% = \frac{1}{10} - \frac{9}{10}$

$$\frac{x \times \frac{6}{5}}{y \times \frac{9}{10}} = \frac{3}{8}$$

$$\frac{x}{y} \times \frac{4}{3} = \frac{3}{8} = \boxed{\frac{x}{y} = \frac{9}{32}}$$

(16) In an examination, A got 25% more marks than B, B got 10% less marks than C, and C got 25% more marks than D, D got 320 out of 500, then A got How many marks?

(a) 405 (b) 450 (c) 360 (d) 400

D C B A

$$\boxed{320 \times \frac{5}{4} \times \frac{9}{10} \times \frac{5}{4}} = \underline{450}$$

Percentage of Percentage

(17) In a big garden 60% of the trees are Coconut tree 25% of the number of Coconut trees are mango tree and 20% of the number of mango trees are apple trees, If the number of apple trees in the garden is 1440, then find the total no. of trees in the garden

(a) 48000 (b) 50,000 (c) 51,000 (d) 45,000

Basic

Total $\longrightarrow 100x$

$\downarrow 60\%$

Coconut $\longrightarrow 60x$

$\downarrow 25\% = \frac{1}{4}$

Mango $\longrightarrow 15x$

$\downarrow 20\% = \frac{1}{5}$

Apple $\longrightarrow \boxed{3x} \longrightarrow 1440$

$100x \longrightarrow \frac{1440 \times 100}{3}$

$$\boxed{x = 48{,}000}$$

~~Adva~~ Advance

$$\underbrace{\underbrace{\underbrace{x \times \frac{3}{5}}_{C} \times \frac{1}{4}}_{M} \times \frac{1}{5}}_{AP} = \overset{480}{\cancel{1440}}$$

$$\frac{x}{100} = 480$$

$$\boxed{x = 48{,}000}$$

(18) ~~The~~ A man spends 40% of his salary on food. 20% on house rent. 10% of entertainment and 10% on Convenience. If his saving at the end of months are. Rs. 1600 then his salary per month is.

(a) 6500 (b) 7500 (c) 8000 (d) 6000

exp $= 40\% + 20\% + 10\% + 10\% = 80\%$

savings $= 20\% \longrightarrow 1600$

Income $= 100\% \longrightarrow \frac{\overset{80}{\cancel{1600}} \times 100}{\cancel{20}}$

Income $= 8000$

Advance

Savings $= 20\% = \frac{1}{5}$

Income $= x$

$$x \times \frac{1}{5} = 1600$$

$$\boxed{x = 8000}$$

(19) out of his total income Mr. Kapur spends 20% on house rent and 70% of the rest of house hold expense. If he save Rs. 7200 what is his total income

(a) 30,000 (b) 35000 (c) 25000 (d) Not

Tree

$I \rightarrow \boxed{100}$

$\downarrow -20\%$

$\boxed{80}$

$\downarrow 70\% = 56$

Sav $\rightarrow \boxed{24} \rightarrow 7200$

$100 = \frac{7200 \times 100}{24}$

$= 30,000$

Advance

$x \times \frac{80}{100} \times \frac{30}{100} = 7200$

Savings

$\boxed{x = 30,000}$

(20) A person spends 25% of his income on house rent, 20% of the remaining part on food and 10% of the remaining part on travelling and finally save 8100 then find his total income

(a) 13,000 (b) 14000 (c) 15000 (d) 16000

$\cancel{x \times \frac{3}{5}}$

$x \times \frac{3}{4} \times \frac{4}{5} \times \frac{9}{10} = 8100$

$\boxed{x = 15,000}$

Election based problems :→

(a) Voter list

↓

Polled vote

↓

Valid vote

↓

Candidate

(21) In an election between two candidates, one got 55% of the valid votes, 20% of the registered votes were invalid. If the total no of votes was 7500, the no of valid votes that the other candidate got was

(a) 2760 (b) 2790 (c) 2700 (d) 2650

1st candidate → 55% of valid

2nd " → 45% " "

Invalid = 20%

polled votes = 7500

$\left(7500 \times \frac{80}{100}\right) \times \frac{45}{100} = \underline{2700}$

valid

(22) In an election between two candidates, 75% of the voters cast their votes, out of which 2% of the votes were declared invalid. A candidate got 9261 votes which were 75% of the total valid votes. Find the total no of voters

(a) 16850 (b) 16300 (c) 16200 (d) 16800

polled → 75%

Invalid → 2%

$\text{Valid} \times \frac{75}{100} = 9261$

$x \times \frac{75}{100} \times \frac{98}{100} \times \frac{75}{100} = 9261$

$\boxed{x = 16,800}$

(23) In an election between two candidates $11\frac{1}{9}\%$ did not cast their votes. $12\frac{1}{2}\%$ cast their votes were declared as invalid. And winner got 60% of the total valid votes and won by 70,000 votes. Find the total no. of voters in the voting list.

(a) 45,5000 (b) 49,000 (c) 45,000 (d) ~~550~~ 50,000

Total → x

Did not vote $= 11\frac{1}{9}\% \rightarrow \frac{1}{9}) \frac{8}{9}$

Invalid $= 12\frac{1}{2}\% \rightarrow \frac{1}{8}) \frac{7}{8}$

Winner $= \boxed{60\%}$ Loser $= \boxed{40\%}$

valid won by $\boxed{20\%} \rightarrow \frac{1}{5}$

$$\boxed{x \times \frac{8}{9} \times \frac{7}{8} \times \frac{1}{5} = 7000.}$$

1000

$\boxed{x = 45,000}$

(24) In an election between two candidate 10% voters did not cast their votes, and 60 votes were declared invalid. The winning candidate scored 47% of the total voters list and he won the election by 308 votes. Find total no. of voters in election.

(a) 6200 (b) 6000 (c) 6500 (d) Not

Voters → valid → $\boxed{\text{Total}}$

Tree method

Total → 100

↓ −10%

Polled → 90%

↓ −60

Valid → 90% − 60

Winner Loser

47%

of valid ~~valid winner~~ → 60. Valid − winner → loser

90% − 60 − 47% → 43% − 60

~~Won~~ won by = winner − looser

47% − (43% − 60) = 308

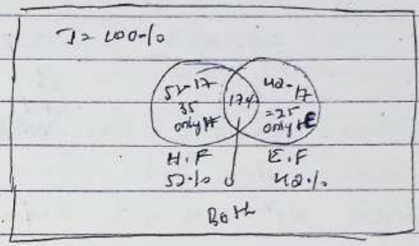
4% + 60 = 308

Set-type Questions:

(25) In an examination 52% students failed in Hindi and 42% in english. If 17% failed in both the subjects. what percentage of students passed in both the subjects.

(a) 38% (b) 33% (c) 23% (d) 30%

~~Data fail~~ Data - fail Set → fail



~~Fail~~ Fail → 35 + 25 + 17 = 77%

pass → 100 − 77 = 23%

Set method = A + B − (A∩B) = A∪B

A → Hindi fail → 52%; B → Eng fail → 42%; (A∩B) → H & E Both fail → 17%; A∪B → H & E Total fail → 77%

52% + 42% − 17% = 77%

pass = 100 − 77 = 23%

(26) A hostel has 800 boys. 75% boys play hockey 50% boys play football. If each boy plays hockey or football or both, then how many boys play both sports?

(a) 100 (b) 200 (c) 400 (d) 600

A + B − (A∩B) = A∪B

A → H.P; B → F.P; (A∩B) → Both H & F; A∪B → Total H & F

75% + 50% − x = 100%

25% = x

$800 \times \frac{1}{4} = 200$

(27) In an examination 60% of the candidates passed in English and 70% of the candidates passed in mathematics but 20% failed in both of these subjects. If 2500 candidates passed in both subjects the no. of who appeared at the examination was

(a) 3500 (b) 5000 (c) 3200 (d) 4000

$A + B - (A \cap B) = A \cup B$

↓ E.P ↓ M.P ↓ both passed ↓ Total passed

$60\% + 70\% - x = 80\%$

$50\% = x$

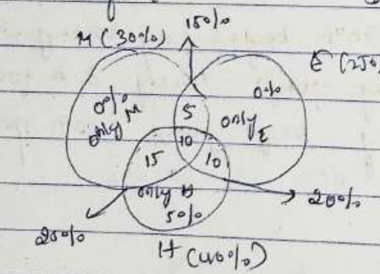
$50\% \to 2500$

$100 \to 5000$

(28) In an examination 30% of the students failed in Math, 25% failed in English, 40% failed in Hindi, if 15% failed in M & E, 20% in E & H, 25% in M & H and 10% " " all 3 then % students who passed in all three subjects.

(a) 50% (b) 45% (c) 55% (d) 65%

value - fail   only - fail



Total failed = $0+5+0+15+10+10+10+5 = 55\%$

Total passed = $100 - 55\% = 45\%$

(29) In a test, it is mandatory for a student to get 33% marks to pass, the student has obtained 25% marks and failed by 40 marks, find the max marks of examination

(a) 400 (b) 500 (c) 600 (d) NOT

Pass = $33\% = 25\% + 40$

$8\% = 40$

$100\% = \frac{40 \times 100}{8} = 500$ marks

(30) A candidate who gets 30% marks in an examination fail by 45 marks, but another candidate who gets 42% gets 45 marks more than the passing marks, find the maximum marks of examination.

(a) 600 (b) 700 (c) 750 (d) 800

$30\% + 45 = 42\% - 45$

$12\% = 90$ marks

$100\% \to \frac{90 \times 100}{12}$ 750

max = 750 marks

(31) In a school, boys and girls are in the ratio 3:2. If 20% of boys and 30% of girls are going to take scholarship, then what is the percentage of students who do not take scholarship?

(a) 50% (b) 72% (c) 56% (d) 76%

Boys : girls

3 : 2 × 10

30 : 20

↓20% ↓30%

taking sch → 6 : 6

Do not " " → 24 : 14 → $\frac{38}{50} \times 100$

$= 76\%$

(32) The ratio of the no. of boys to that girls in a village is 3:2. If 30% boys and 70% girls appeared in an examination. the ratio of the no. students appeared in the examination to that donot appeared in the same examination

(a) 23:27 (b) 27:25 (c) 21:25 (d) 25:21

Boys : Girls
30 : 20
↓30% ↓70%
Appeared 9 : 14 → 23
do not appear 21 : 6 → 27

(33). The population of a town is 6000. If males are increased by 5% and female are increased by 9% then population will become 6500 after 1 yr. Find the no. of males and females

(a) M = 1000 F = 5000 (b) 5000, 1000 (c) 2000 4000 (d) 4000 2000

Traditional (M = x F = y)

$x + y = 6000$ — (i)

$\frac{x \times 105}{100} + \frac{y \times 109}{100} = 6500$ — (ii)

Extra $\frac{x \times 5}{100} + \frac{y \times 9}{100} = 500$ → (iii)

Logic:

$6500 - 6000$

Total = 6000 → 300 but difference = 500

males = $\boxed{5\%}$

Female = $\boxed{5\%}$ + 4%

200 = 4% of female

$\frac{200 \times 100}{4}$ ← 100%

= 5000 = F

M = 1000 ✓

(34). A Company gives 10% commission to his saleman upto the sale of 8000 Rs and a Commission of 15% on the sales above Rs. 8000. If the salesman deposited Rs. 42130. in the company after deduction his commission. Find total sales.

(a) Rs. 50,000 (b) 45000 Rs (c) Rs. = 55,000 (d) NOT

Total sales = 100%

10% | 8000 | 15% | 8000+

$\boxed{15\%} + \boxed{4\%}$ $\boxed{15\%}$ → for above 8000+

For 8000 → $\boxed{4\%}$

↓

4% in for 8000 $\frac{8000 \times 4}{100}$

= 320

Total sales → deduction + Given

15% + 320 + 42180 = 100%

85% → 42500

100% → $\frac{42500 \times 100}{85}$

= 55,000

(35) A, B, C, D purchased a Cine-multiplex for Rs. 56 lakhs. The contribution of B, C, D = 460% of A, alone, The contribution of A, C, D = 366.66% of B. and contri-of C = 40% of A, B, D. Then the amount contributed by D is

(a) 10 lakh (b) 16 lakh (c) 12 lakh (d) 18 lakh

A = 100 B+C+D → 460 — $\frac{23}{5}$

B = 100 A+C+D → (300+66.66)%

$3 + \frac{2}{3}$

$\frac{11}{3}$ → A+C+D, → B

C = 100 A+B+D → 40 : $\frac{2}{5}$

A = 5 + B+C+D = 23 = 28×1

B = 3 + A+C+D = 11 = 14×2

C = 2 + A+B+D = 5 = 7×4

A = 5 B = 3×2 C = 2×4

A = 5 B = 6 C = 8

A+B+C+D = 28

19+D = 28

D = 9

28 parts = 56 lakh

9 → $\frac{56 \times 9}{28}$

D = 18 lakh

Ratio:→ A : B

weight $\boxed{7} : \boxed{8}$

$\boxed{14} : \boxed{16}$ $A = 7x$ $\boxed{1\text{ part} = x}$

$\boxed{21} : \boxed{24}$ $B = 8x$

proportion:→

$a : b :: c : d$

$2 \quad 3 = 2 : 3$

$a : b :: c : d$

$\frac{a}{b} = \frac{c}{d}$

$\boxed{a \times d = b \times c}$

Fourth proportion = $a : b :: c : ?$

3rd proportion = $a : b : c$

$a : b :: b : c$

$a \times c = b \times b$

$\boxed{c = \frac{b^2}{a}}$

Ex:→ $9 : 15 : ?$ $c = \frac{b^2}{a}$

$? = \frac{15 \times 15}{9} = 25$

Mean proportion:→ $a : b$

$a : x : b$

$a : x :: x : b$

$x^2 = ab$

$x = \sqrt{ab}$

Ex:- $9 : 25$

$x = \sqrt{ab}$

$x = \sqrt{9 \times 25}$

$= 3 \times 5$

$x = 15$

Ratio :→ $a : b$

Duplicate ratio :→ $a^2 : b^2$

Sub duplicate ratio :→ $\sqrt{a} : \sqrt{b}$

Triplicate ratio :→ $a^3 : b^3$

Sub triplicate ratio :→ $\sqrt[3]{a} : \sqrt[3]{b}$

① Rs. 15,500 divided into three parts proportional to $\frac{1}{2} : \frac{1}{3} : \frac{1}{5}$ are respectively?

(a) Rs. 7500, Rs. 5000, Rs. 3000

(b) Rs. 5000, Rs. 7500, Rs. 3000

(c) Rs. 3000, Rs. 5000, Rs. 7500

(d) NOT

15,550

$\left(\frac{1}{2} : \frac{1}{3} : \frac{1}{5}\right) \times 30$

$15 : 10 : 6$ = Total = 31 parts

$\frac{15,500}{31} : \frac{15,500}{31} : \frac{15,000}{31}$

$500 \times 15 : 500 \times 10 : 500 \times 6$

$7500 : 5000 : 3000$

② The ratio of expenses and savings of a person is 26:3 If your monthly income is Rs. 145000 then what is his monthly savings?

(a) Rs. ₹580 (b) Rs. 700 (c) Rs. 1500 (d) Rs. 1550

$E + d = 2$

$26 : \boxed{3} = \boxed{29}$

$\downarrow \quad \downarrow$

$? \quad 14,500$

$? = \frac{14,500 \times 3}{29}$

$? = 1500$

③ The monthly income of A and B is Rs. 6000 and Rs. 8000 respectively. If their expenses are 70% and 80% respectively, find the ratio of their savings of both

(a) 9:7 (b) 8:9 (c) 5:8 (d) 7:7

	A	B
$I \rightarrow$	6000	8000
$E \rightarrow$	70%	80%

$\left[S \rightarrow \frac{6000 \times 30}{100} \rightarrow 30\% : \frac{8000 \times 20}{100} \right] \rightarrow$ Approach

$900 : 800$

$9 : 7$

④ A money is divided into A, B and C in the ratio of 3:5:7. If C's share is Rs. 1600 more than A's share then what is B's share?

(a) 1600 (b) 2000 (c) 1500 (d) 1000

A : B : C

3 : 5 : 7

4 parts $\longrightarrow$ 1600

B's = 5 parts $\rightarrow \frac{1600 \times 5}{4}$

$\boxed{B's = 2000}$

Uniform ratio: $\longrightarrow$

If $A:B = (2:3)$ if $\times 4$ the $8:12$

$B:C = (4:5)$ $\times 3$ $12:15$

$A:B:C = 8:12:15$

A	B	C
2	3	○
○	4	5

$\longrightarrow$

A	B	C
2	3	3
4	4	5
8	12	15

$A:B = 2:3$

$B:C = 4:5$

$C:D = 3:5$

$D:E = 4:3$

A	B	C	D	E
2	3	○	○	○
○	4	5	○	○
○	○	3	5	○
○	○	○	4	3

$\longrightarrow$

A	B	C	D	E
2	3	3	3	3
4	4	5	5	5
3	3	3	5	5
×4	×4	×4	×4	×3
32	48	60	100	75

⑤ A sum of money is distributed among A, B & C such that B gets 12.5% more than A, but 25% more than C. If A & C is Rs. 5700. Find the difference of A & C

(a) 200Rs (b) 300Rs (c) 400Rs (d) 500Rs

$12.5 : \frac{1}{8} \quad 25 : \frac{1}{4}$

$B = \frac{1}{8}A$

$\frac{A}{B} = \frac{8}{9}$

A	B	C
8	9	9
5	5	4
40	45	36

$A\&C = 40+36 = 76$ part $\longrightarrow 5700$

$A\&C$ diff $= 4$ part $\longrightarrow \frac{5700 \times 4}{76}$

$\cancel{400}$ $\underline{300}$ → diff

(6) A sum of money is distributed among A, B and C such that B gets 10% less than A, but 20% less than C and D gets $33\frac{1}{3}$% more than C. Then find D is that what % more than A

(a) 45% (b) 50% (c) 60% (d) NOT

$B = \frac{1}{10}$

A : B : C : D
10 : 9 : 9 : 9 3
4 : 4 : 5 : 5
3 : 3 : 3 : 4
40 : 36 : 45 : 60

$\% = \frac{20}{40} \times 100 = 50\%$

Alternate (compounded ratio)

A : B = 10 : 9
B : C = 4 : 5
C : D = 3 : 4

$\frac{A}{B} \times \frac{B}{C} \times \frac{C}{D} = \frac{A}{D} = A : D$

$\frac{10}{3} \times \frac{4}{5} \times \frac{3}{4} = \frac{2}{3}$ 2 : 3

more % $\frac{1}{2} \times 100 = 50\%$

(7) A person distributes Rs. 3,30,000 to his daughter, wife and son in such a way that ½ part of the daughter is ¼ th part of the wife and ⅕ part of the son is equal. How much is the daughter's share?

(a) Rs. 30,000 (b) Rs. 60,000 (c) 90,000 Rs (d) Rs. 1,20,000

D : W : S

$\left(\frac{1}{2}D = \frac{1}{4}W = \frac{1}{5}S\right) = K$

Trick → Eqn → $\frac{1}{2} = \frac{1}{4} = \frac{1}{5}$

Reciprocal

D : W : S
2K : 4K : 5K
$\boxed{2 : 4 : 5}$

$\boxed{2 : 4 : 5}$

$\frac{3,30,000}{11} \times 2$ (30000)

= 60,000

(8) Two numbers are in ratio of 7:13. When 24 is added in each number then ratio becomes 11:17. Find difference between both initial numbers

(a) 24 (b) 42 (c) 36 (d) 60 (e) Not

Traditional :-

1st → 7x 13x
↓24 ↓24
(7x+24) : (13x+24)
11 : 17

$\frac{7x+24}{13x+24} = \frac{11}{17}$

$\Rightarrow \boxed{x = 1}$

Advance

$\begin{array}{ccc} 7 & : & 13 \\ 11 & : & 17 \\ 4 & : & 4p \rightarrow 24 \end{array}$

13 − 7 6p → $\frac{24 \times 6}{4}$

diff = 36

(9) The ratio of monthly incomes of Ram and Shyam is 9:8 and their monthly expenditures are in the ratio 7:6. If each of them saves Rs. 8000 per month; find the sum of their monthly incomes.

(a) 58000 (b) 68,000 (c) 56,000 (d) 60,000 (e) NOT

R : S
I → 9 : 8
E → 7 : 6
S → 2 : 2p → 8000

17p → $\frac{8000 \times 17}{2}$ = 68,000

Traditional

9x − 7y = 8000
8x − 6y = 8000

Exp = $\frac{7x + 8000}{6x + 8000} = \frac{9}{8}$

(10) The ratio of milk to water is 7:9. If 27 litres of water is added, the ratio becomes 7:12 then find the amount of milk in the mixture

(a) 27 litres (b) 9 ltr (c) 81 ltr (d) 81 ltr (e) 45 ltr

M : W
7 : 9
7 : 12
0 : 3p → 27 litre

p = 9 litres

Milk = 7 × 9 = 63

(11) The ratio of milk to water is 5:3. If 63 ltrs of that is added the ratio becomes 7:11, then find the new amount of water in the mixture

(a) 120 ltrs (b) 160 ltrs (c) 180 ltrs (d) 110 ltrs (e) 170 ltrs

M : W
(5 : 3) = 8 × 7
(7 : 11) = ~~28~~ × 5

water is added since milk remains constant to make milk ratio equal

M : W
35 : 21
35 : 55

34p → 68
55p → $\frac{68 \times 55}{34}$ = 110 litres

(12) The respective ratio between the A's age after 3 yrs and B's age after 3 yrs ago is 10:9 and the respective ratio between the A's age 3 yrs ago and B's age after 3 yrs is 17:21. What is A's present age?

(a) 39 (b) 35 (c) 27 (d) 41 (e) 37

present age; A & B

$\frac{A+3}{B+3} = \frac{10}{9}$ $\frac{A-3}{B+3} = \frac{17}{21}$

& substituting A value in equations verify & solve equations

(13) Rs 5200 was to be divided among A, B and C in the ratio of $\frac{1}{3} : \frac{1}{4} : \frac{1}{2}$, but by mistake it was divided in the ratio 7:4:15. The amount in excess received by C was

(a) Rs. 550 (b) Rs. 600 (c) Rs. 450 (d) Rs. 750

A : B : C
($\frac{1}{3} : \frac{1}{4} : \frac{1}{2}$) × 12
4 : 3 : 6 = 13 on balancing × 2
7 : 4 : 15 = 26 × 1
= 8 : 6 : 12
7 : 4 : 15

26 parts → 5200
2 p → $\frac{5200 \times 2}{26}$

$\boxed{\text{C's excess} = 600}$

(14) A sum of money was to be divided among A, B & C in the ratio of $\frac{1}{4} : \frac{1}{6} : \frac{1}{3}$, but by mistake it was divided in the ratio 7:8:12. If B received Rs. 2000 excess then find that money was distributed among A, B and C

(a) Rs. 5400 (b) Rs. 2700 (c) Rs. 2400 (d) NOT.

A : B : C
($\frac{1}{4} : \frac{1}{6} : \frac{1}{3}$) × 12
3 : 2 : 4 = 9 × 3
7 : 8 : 12 = 27 × 1

9 : 6 : 12
7 : 8 : 12

2p → 2000
27 → $\frac{2000 \times 27}{2}$ = 1000 × 27 = 2700

(15) The income ratio of A, B and C is 4:5:7 and expenditure ratio is 5:4:6 respectively. If "A" saves 25% of his income but C saves Rs. 6800. Then find savings of B.

(a) Rs. 5200 (b) Rs. 1300 (c) 2600 (d) 3900

A : B : C
in → 4 : 5 : 7 (x)
ex → 5 : 4 : 6 (y)
sav → 1/4 :

A saves 25% = $\frac{1}{4} \frac{\to s}{\to I}$

4x − 5y = x
3x = 5y
$\frac{x}{y} = \frac{5}{3}$

A : B : C
4 : 5 : 7 = 16 → I(x)
5 : 4 : 6 = 15 → E(y)

4×5 : 5×5 : 7×5
5×3 : 4×3 : 6×3

20 : 25 : 35
15 : 12 : 18

17p → 6800
13p = $\frac{6800 \times 13}{17}$

(16) The expenditure ratio of A:B:C is 4:9:5 If A, B and C save 20%, 25% and 16.66% of their income respectively. If the total income of A, B, C together is Rs 4600. Find the income difference of B and C

(a) Rs. 1000 (b) Rs. 1200 (c) Rs. 1300 (d) NOT.

A : B : C

EX → 4 : 9 : 5

$\frac{1}{4}$: $\frac{1}{4}$ = 3 : $\frac{1}{6}$

5 : 12 : 6

23 parts → 4600

6 part → $\frac{4600}{23}$ = 1200

(17) 3rd division in 5:6:7 respectively. After recheck the Copy, 200 students promoted from 3rd division to 1st and 2nd division and now ratio becomes 4:5:3 respectively. Then find the total number of students in the Class?

(a) 360 (b) 720 (c) 1080 (d) 1440 (e) NOT

1st : 2nd : 3rd

5 : 6 : 7 = 18 × 2 = 36

4 : 5 : 3 = 12 × 3 = 36

10 : 12 : 14

12 : 15 : 9

5p → 200

36 → $\frac{200 \times 36}{5}$

$\boxed{\text{Total} = 1440}$

Mixture problems:-

(18) In a 200 gm alloy, the ratio of copper and zinc is 2:3, how much copper is added to it and the ratio of copper and zinc in the new alloy is 3:2?

(a) 100 gm (b) 200 gm (c) 120 (d) 150

Cu : Zn

2 : 3

$\boxed{80}$: $\boxed{120}$

$\boxed{180}$ (120) → no change

3 : 2

2 part → 120

1 part → 60

(19) A and B are alloys of gold and copper in the ratio 7:2 and 7:11 respectively. If equal quantities of these two alloys are melted to form new alloy C, then the ratio of gold and copper in C is?

(a) 7:5 (b) 5:7 (c) 4:5 (d) 5:4

A: G : C, (7 : 2)×2, 9, 14 : 4

B: G : C, (7 : 11)×1, 18, 7 : 11

21 : 15

7 : 5

(20) Two mixture A and B contain 80 ltr and 60 ltr mixture of milk and water in the ratio 5:3 and 7:5 respectively. If both mixture are poured in new vessel C then what will be the ratio between milk & water?

(a) 17:11 (b) 15:11 (c) 11:17 (d) NOT

A (80): M : W, 5 : 3, 50 : 30

B (60): M : W, 7 : 5, 35 : 25

85 : 55

17 : 11

(21) Acid and water are mixed in a vessel A in the ratio of 5:2 and in the vessel B in the ratio 8:5. In what proportion should quantities be taken out from two vessels so as to form a mixture in which the acid and H_2O will be in the ratio 9:4

(a) 7:2 (b) 2:7 (c) 7:4 (d) 2:3

A (7x): A : W, 5x : 2x

B (13y): A : W, 8y : 5y

$\frac{A}{W} = \frac{5x+8y}{2x+5y} = \frac{9}{4}$

20x + 32y = 18x + 45y

$\frac{x}{y} = \frac{13}{2}$

7x : 13y

7×13 : 13×2

$\boxed{7:2}$

A : W

9 : 4

(22) Three containers A, B and C having mixtures of milk and water in the ratio 1:5, 3:5, 5:7, respectively. If the capacities of the containers are in the ratio 5:4:5. find the ratio of milk and water, if the mixtures of all three containers are mixed together

(a) 63:121 (b) 53:115 (c) 77:133 (d) 115:53 (e) NOT

A (5)×24 → 120 B (4)×24 → 96 C (5)×24 → 120

M : W M : W M : W

1 : 5 3 : 5 5 : 7

6 8 12

24

20 : 100 36 : 60 50 : 70

106 : 230

53 : 115

(23) Fresh fruit contains 80% water and dry fruit contains 15% water. How many kg's dry fruits can be obtained out of 255 kg fresh fruits?

(a) 50kg (b) 60kg (c) 65kg (d) 80kg

Fresh fruit

255 kg

W F

80% 20%

4 : 1

204 : 51

(9) 51

15% : 85%

3 : 17

17 → 51

Dry → 60kg

(24) In an office of 1200 employees, the ratio of urban to rural members of staff 8:7. After joining of some new employees, out of which 20 are rural, the ratio becomes 5:4. The no. of new urban employees is,

(a) 100 (b) 85 (c) 76 (d) 107 (e) NOT

1200

8 7

$\frac{1200}{15}$ 80

(85) 640 : 560

+20

New urban

725 : 580

5 : 4

⟹ 4p → 580

5p → $\frac{580 \times 5}{4}$ = 725

(25) The ratio of students in a coaching preparing for B.Tech. and MBA is 4:5. The ratio of fees collected from each of B.Tech and MBA students is 25:16. If the total amount collected from all the students is 1.62 lakh, what is the total amount collected from only MBA aspirants?

(a) 62K (b) 80K (c) 72K (d) NOT

	B.Tech		MBA
S →	4	:	5
F →	25	:	16
	100	:	80
	5	:	4

→ 9 → 1,62,000

4p → $\frac{1,62,000 \times 4}{9}$ 18000×4

→ MBA = 72,000

(26) Swami bought pulses of worth Rs 32/kg and Rs. 45/kg he mixed them with a third variety in the ratio 1:1:2. If the mixture is worth Rs 88/kg. then the price of the third variety per kg will be

(a) 169.50 (b) 175.50 (c) 137.50 (d) 145.50 (e) NOT

1st	2nd	3rd
32/kg	45/kg	x/kg
1 :	1 :	2

32 + 45 + 2x = 88×4

77 + 2x = 352

$\boxed{x = 137.5}$

Coins Based

(27) A bag contains a certain number of coins of Rs 1, 50 paise and 25 paise in ratio 7:8:4. If there is Rs 360 in the bag then calculate no. of coins of 25 paise

(a) 210 (b) 120 (c) 240 (d) 360 (e) 250

	1 Rs	50p	25p
No's →	7x :	8x :	4x
Value →	7x :	$8x/2$:	$4x/4$

$\hookrightarrow 7x + 4x + x \to 360$

$12x = 360$

$\boxed{x = 30}$ $25p \to 4x \to 4 \times 30 = 120$

(28) A box contains 280 coins of Rs 1, 50 paise and 25 paise. The value of each kind of coin are in the ratio of 8:4:3. The calculate number of coins of 50 paise

(a) 80 (b) 120 (c) 60 (d) 100 (e) 250

	₹1	₹50	₹25
Value →	8x :	4x :	3x
No's →	8x :	4x×2 :	3x×4

$\therefore 8x + 8x + 12x = 280$

$28x = 280$

$\boxed{x = 10}$ $\boxed{8x = 80}$ No's

(29) One year ago the ratio between A's and B's salary was 3:4. The ratio of their individual salaries between last year's and this year's salaries are 4:5 and 2:3 respectively. At present the total of their salary is 4160. The salary of A now is?

(a) 1600 (b) 1700 (c) 1800 (d) 1900 (e) NOT

$\frac{A_{-1}}{B_{-1}} = \frac{3}{4}$ $\frac{A_{-1}}{A_p} = \frac{4}{5}$ $\frac{B_{-1}}{B_p} = \frac{2}{3}$

$A_p + B_p = 4160$

$\frac{A_{-1}}{B_{-1}} \times \frac{A_p}{A_{-1}} \times \frac{B_{-1}}{B_p} = \frac{A_p}{B_p}$

$\frac{3}{4} \times \frac{5}{4} \times \frac{2}{3} = \frac{5}{8}$ $\Longrightarrow \frac{A_p}{B_p} = \frac{5}{8}$

$13p = 4160$

Ans $5p = \frac{4160 \times 5}{13} = \boxed{1600}$

(30) If the ratio of incomes of A and B in 2001 is 2:3 and the ratio of incomes of A in 2001 and 2002 is 4:5. Find the expenditure of A in 2002. If saving in the same year is Rs 4000. It is given that in 2001, the sum of income of A and B is Rs. 25,000

(a) ₹5000 (b) ₹9500 (c) ₹8500 (d) ₹10,5000 (e) ₹3500

2001 → Income → A : B (25000)

2 : 3

10,000

Income → A → 2001 : 2002

$2p = 2500$ 4 : 5×2500

$\boxed{10,000}$: 125000

2002 → A → exp → ?

$\hookrightarrow$ savings → $\boxed{4000}$

$\hookrightarrow$ Income → $\boxed{12,500}$ → 12,500 − 4000

$\boxed{8,500}$ → exp

PROFIT, LOSS AND DISCOUNT :—

CP = Cost price (100) → SP = selling price (105) ← MP = marked price (150)

+ markup% (CP → MP)

+ profit % (CP → SP)

− discount% (MP → SP)

profit = (SP > CP)   loss (CP > SP)   Discount (SP < MP)

profit = SP − CP   loss = CP − SP   Discount = MP − SP

$$\text{profit\%} = \frac{\text{profit}}{CP} \times 100 \qquad \text{loss\%} = \frac{\text{loss}}{CP} \times 100 \qquad \text{Discount} = \frac{\text{Discount}}{MP} \times 100$$

$= \frac{5}{100} \times 100$ $= \frac{45}{150} \times 100$

$= 5\%$ $= 30\%$

Basics :—

CP = 4000 (100%), profit = 23% (23%), S.P = $4000 \times \frac{123}{100} = 4920$ (123%)

CP = 3000 (100%), loss = 37% (37%), SP = $3000 \times \frac{63}{100} = 1890$ (63%)

CP = 840, profit = $16\frac{2}{3}\%$ $= \frac{1}{6}$, SP = $840 \times \frac{7}{6} = 980$

CP = 640, loss = 12.5% $= \frac{1}{8}$ $= 8-1$ $= \frac{7}{8}$, SP = $640 \times \frac{7}{8} = 560$

SP = 980, profit = $16\frac{2}{3}\%$ (1 → P, 6 → CP), $CP \times \frac{7}{6} = SP$, $CP \times \frac{7}{6} = 980$, $\boxed{CP = 840}$

S.P = 560, loss = 12.5% $\left(\frac{1}{8}\right)$, $CP \times \frac{7}{8} = SP$, $CP \times \frac{7}{8} = 560$, $\boxed{CP = 640}$

SP = 4920, profit = 23%, $CP \times \frac{123}{100} = SP$, $CP \times \frac{123}{100} = 4920$, $\boxed{CP = 4000}$

(1) A sold some article to B at 12.5% profit. B sold to C at 20% loss and C sold it to D at 16.66% loss. If D bought it at Rs. 2700. Then find cost of article for A in Rs.

(a) 4000 (b) 3500 (c) 3600 (d) 4200 (e) Not

A → B → C → D

$CP \times \frac{9}{8} \times \frac{4}{5} \times \frac{5}{6} = 2700$ (S.P of AB)

$\boxed{CP = 3600}$

12.5% $= \frac{1}{8}$

20% $= \frac{1}{5}$

16.66% $= \frac{1}{6}$

(2) A buys a scooter for Rs. 6800 and spends at Rs. 1200 on its repairs. If he sells the scooter for 9200 his gain % is

(a) 13% (b) 14% (c) 12.5% (d) 16% (e) 15%

c.p + exp

6800 + 1200 = $\boxed{8000} \to$ FCP

s.p = 9200

profit = $\frac{1200}{8000} \times 100$

$\boxed{\text{profit\%} = 15\%}$

③ 10 kg Rice was bought at the rate of the Rs. 10/kg and 15 kg rice was bought at the rate of Rs. 12/kg. Find the % profit / loss, if the total rice is sold for Rs. 16/kg

ⓐ 35% ⓑ 40% ⓒ $42\frac{6}{7}\%$ ⓓ $57\frac{1}{7}\%$ ⓔ NOT

10 kg × 10 → 100

15 kg × 12 → 180

280

$\frac{280}{25} = \frac{56}{5}$/kg

C.P : S.P

$\frac{56}{5}$: 16

7 : 10 profit = 3 profit% = $\frac{3}{7} \times 100 = \frac{1}{7} \times 3 \times 100$

$= 14\frac{2}{7}\% \times 3$

$= 42\frac{6}{7}\%$

④ A person sold an article at Rs. 535 loss incurred will be equal to the profit incurred when an article is sold at Rs. 765. At what price it should sell to get 10% profit

ⓐ ₹ 600 ⓑ ₹ 650 ⓒ ₹ 390 ⓓ ₹ 540 ⓔ NOT

$535 \xleftarrow{-x} \boxed{cp} \xrightarrow{+x} 765$

C.p + x = 765

Cp − x = 535

2cp = 1300

$\boxed{cp = 650}$ $\boxed{650 \times \frac{110}{100} = 715}$

⑤ A person sold an article at Rs. 1840 and profit incurred will be 12.5% more than and the loss incurred when article is sold at Rs. 1500. What price it should sell to get 20% profit in Rs.

ⓐ 1992 ⓑ 1994 ⓒ 1995 ⓓ 1996 ⓔ NOT

12.5% = $\frac{1}{8}$

$\boxed{1500} \xleftarrow[\text{loss}]{8x} \boxed{Cp} \xrightarrow[\text{profit}]{9x} \boxed{1840}$ → difference

17x = 340

$\boxed{x = 20}$

cp + 9x = 1840

cp − 8x = 1500

17x = 340

$\boxed{x = 20}$

C.P $\xrightarrow{9x}$ 1840

9×20 = 180

1840 − 180 = 1660

C.P = 1660

20% profit = $1660 \times \frac{6}{5} = \boxed{1992}$

⑥ A person sells 5 cows, 1st at a profit of 6% 2nd at 2% loss, 3rd sold at 12% profit, 4th on 8% profit, what he should sell 5th cow to gain overall 10% profit?

ⓐ 25 ⓑ 26 ⓒ 30 ⓓ 35 ⓔ 20

cp = $\boxed{100}$ → 106, 98, 112, 108 = 424

×5 → $\boxed{500}$ → +10% → $\boxed{550}$

$\boxed{126}$

126 − 100

26%

Quantity Based Question :→

① A shopkeeper bought 20 pens and sold 15 pens for same price. What had gained or cost?

ⓐ 25% + ⓑ 33% − ⓒ 10% − ⓓ 20% − ⓔ 40% +

$20\, s.c.p = 15 \times s.p$

$\frac{c.p}{s.p} = \frac{15}{20} = \frac{3}{4}$ profit% $= \frac{1}{3} \times 100$

$= 33.33\% \;(+)$

(8). Cost price of 72 articles is equal to selling price of 120 articles. Find the profit or loss %

(a) 20% (−) (b) 40% (−) (c) 20% (+) (d) 40% (+)

$72\, c.p = 120\, s.p$

$\frac{c.p}{s.p} = \frac{120}{72} = \frac{5}{3}$ loss% $= \frac{2}{5} \times 100$

$\boxed{\text{loss} = 40\%}$

(9) By selling 25m of cloth a trader gains selling price of 5m cloth. The gain of trader in % is

(a) 25% (b) 20% (c) 27% (d) 29%

$s.p = c.p + \text{profit}$

$25\, s.p = 25\, c.p + 5\, s.p$

$20\, s.p = 25\, c.p$; $\frac{c.p}{s.p} = \frac{4}{5}$ p% $= \frac{1}{4} \times 100$

$= 25\%$

(10) On selling 17 balls at ₹ 720, there is a loss equal to cost price of 5 balls. The cp of ball is

(a) 45 (b) 50 (c) 55 (d) 60

$s.p = c.p - \text{loss}$

$17\, s.p = 17\, c.p - 5\, c.p$

$17\, s.p = 12\, c.p$; $\frac{c.p}{s.p} = \frac{17}{12}$ loss% $= \frac{5}{17} \times 100$

s.p of 12 = 720

$17 \to \frac{720 \times 17}{12}$

17 balls = 60×17

1 ball = ...

(11) Ravi bought some toffees at the rate of 27 toffees for Rs. 40 and sold them at the rate of 17 toffees for 32. His profit % will be?

(a) 25% (b) 20% (c) 30% (d) 15%

LCM method (27×18 = 54)

(27 → 40) × 2

54 → 80

(17 → 32) × 3 16+

54 → 96

profit% $= \frac{16}{80} \times 100$

$= 20\%$

c.p method & s.p method

c.p = 27 toffees → 40

1 toffee → $\frac{40}{27}$

s.p = 17 toffee → 32

1 toffee → $\frac{32}{17}$

c.p : s.p

$\frac{40}{27} : \frac{32}{17}$

5 : 6 → profit $= \frac{1}{5} \times 100$

$\boxed{p\% = 20\%}$

(12) A man purchased some eggs at 3 for Rs.5 and sold them at 5 for Rs.12. Thus he gained Rs.143 at all. The no. of eggs he bought is

(a) 210 (b) 200 (c) 195 (d) 190

c.p: 3 eggs → 5 ×5 = 15 eggs = 25

s.p: 5 eggs → 12 ×3 = 15 eggs = 36 — 11

profit 11 parts → 143 ; 1 part = 13

c.p 25 parts → $\frac{143 \times 25}{11} = 25 \times 13$

₹5 → 3 eggs

1 Rs → $\frac{3}{5}$ eggs

$25 \times 13 \to \frac{3}{5} \times 25 \times 13$

$\boxed{= 195 \text{ eggs}}$

(13) An egg seller sells his eggs only in the packs of 3 eggs, 6 eggs, 9 eggs, 12 eggs etc. but the rate is not necessary uniform. One day Raju (which is not

the (same egg seller) purchased at the rate of 3 eggs for a rupee and next hour he purchased equal no of eggs at the eggs at the rate of 6 eggs for a rupee. Next day he sold all the eggs at the rate of 9 eggs for ₹2. what is % + or - ?

(a) 10% - (b) 11.11% - (c) 30% - (d) 2.5% +

3 eggs → 1/- → 18 eggs → 6
6 eggs → 1/- → 18 eggs → 3   ] 9

LCM of 3, 6, 9 = 18

sells → 9 eggs → 2/- ×4
36 eggs → 8

loss

$$\text{loss}\% = \frac{1}{9} \times 100 = 11.11\%$$

Profit % calculated on s.p

(19) A shopkeeper sells his article at $16\frac{2}{3}\%$ profit on s.p. find his actual profit %

(a) 10% (b) 12.5% (c) 20% (d) 25% (e) 33.33%

profit = $16\frac{2}{3}\% = \frac{1 \to P}{6 \to cp}$ but $\frac{1 \to \text{profit}}{6 \to s.p}$ based on qstn

s.p = 6   p = 1   cp = 5

$$p\% = \frac{1}{5} \times 100 = 20\%$$

(20) Mohit calculates his profit % on the s.p where as Sonu calculates on her cp. They find that the difference of their profits is ₹195. If the selling price is same for both and mohit earned 25% profit where as Sonu earned 30% profit. find the c.p of Mohit in Rs.

(a) 7605 (b) 6315 (c) 6405 (d) 7505 (e) 7605

Mohit (sp) → $25\% = \frac{1 \to \text{profit}}{4 \to sp}$ sp = 4   p = 1   cp = 3

Sonu (cp) → $30\% = \frac{3 \to P}{10 \to cp}$ s.p = 13   p = 3   cp = 10

	c.p	s.p	
M →	3	4	×13 In question s.p of both are same hence equal them
S →	10	13	×4
M →	39	52	→ p = 13
S →	40	52	→ p = 12

In question → 1 part = 195
39 → 195 × 39 = 7605

(16) Alligation based problems

(16) Sanjay buys some articles for Rs. 1,80,000. He sells 2/5th of it at a loss of 12%. If he wants to earn overall profit of 18% on selling all the articles, then at what profit % he sells the remaining articles?

(a) 48% (b) 73% (c) 42% (d) 38% (e) 60%

Base = ~~1,80,000~~

(parts) Total → $\boxed{5}$ ... 1st = 2   2nd = 3

let cp = 1 = 100

5 → $\boxed{100}$ → 500

loss on 1st part profit on part = 2 → $\boxed{200}$ → $200 \times \frac{88}{100}$ = 176

profit on overall = 500 → 118% → $\frac{500 \times 118}{100}$ = 590

590 − 176 = $\boxed{414}$ → 2nd part s.p

c.p 3 → 300
s.p 3 → 414

$$\frac{114}{300} \times 100$$

$\boxed{p\% = 38\%}$

Alligation method

Total = 5 $\quad$ 1st [2] $\quad$ 2nd [3]

1st		2nd
$\boxed{-12\%}$		$\boxed{x\%}$
	overall $\boxed{+18\%}$	
2	:	3
$x-18$	:	$18-(-12)$
$x-18$	:	30

$\frac{x-18}{30} = \frac{2}{3} \Rightarrow x = 38$

(17) A person buys 240 articles @ Rs. 2700 If sells 4/9th part of all particles at 30% loss. At what profit % he has to sell remaining articles so that he will get neither profit nor loss in all over the business.

(a) 20% (b) 24% (c) 28% (d) 32% (e) 36%

$\frac{4}{9}$		$\frac{5}{9}$
1st		2nd
-30%		$x\%$
	overall 0%	
$x-0$		$0-(-30\%)$
4	:	5

$\frac{x}{30_6} = \frac{4}{5}$

$\boxed{x = 24\%}$

(18) A person bought a horse and a vehicle for Rs. 20,000. He sold the horse at 20% profit and a vehicle at 10% loss, thus he got a profit 2% in the total transaction. What is the purchase price of the horse

(a) ₹ 7100 (b) ₹ 7500 (c) ₹ 8000 (d) ₹ 9000 (e) NOT

~~Hose~~ Horse		vehicle
20%		-10%
	overall 2%	
$2-(-10)$		$20-2$
$\cancel{12}$ 2	:	$\cancel{18}$ 3

$\to$ 5part = 20,000

2part = $\frac{\cancel{20,000}^{4000} \times 2}{\cancel{5}}$

$= 8,000$

(19) A man purchased 5 horses and 10 cows for ₹ 10,000. He sells the horses @ 15% + and cows @ 10% −. Thus he gets ₹ 375 as profit. Find the cost of 1 horse

(a) 450
(b) 1100
(c) 800
(d) NOT

profit % $= \frac{375}{10,000} \times 100$

$= 3.75$

Horse		Cows
$+15\%$		-10%
	overall 3.75	
11.25	:	13.75
45	:	55
9	:	11

$\to$ 20parts $\to$ 10,000

11 $\to$ $\frac{\cancel{10,000}^{500} \times 11}{\cancel{20}}$

5horse = 5500

1 horse = $\frac{5500}{5} = \boxed{1100}$

(20) ~~A man purchased 5 horses~~ In what ratio must rice @ ₹ 31 per kg be mixed with rice @ ₹ 36 per kg, so that the mixture worth ₹ 32.50 a kg?

(a) 5:3 (b) 3:2 (c) 7:3 (d) 5:7 (e) NOT

Rice 1		Rice 2
31		36
	overall 32.50	
3.5		1.50
7	:	3

(21) A trader sells two brands of petrol; one is Extra premium (EP) and other one is 'speed' (SP). He mixes 12 ltrs of EP with 3 litres of speed and by selling this mixture at the price of EP he gain the profit of 9.09%. If the price of (EP) is ₹ 48 per litre, then the price of speed is

(a) ₹ 38 (b) ₹ 28 (c) ₹ 44 (d) ₹ 42 (e) NOT

(EP) s.p = 48/ltr profit = 9.09 = $\frac{1}{11}$ cp = 11 p = 1 sp = 12

12 → 48
$1p = \boxed{4}$
hence cp = 11p = 11×4
= 44 per ltr

E.P cp = 48/ltr   SP cp = x/ltr

Mix cp = 44/ltr

44−x : 4
12 : 3
4 : 1

$\frac{44-x}{4} = \frac{4}{1}$

44 − x = 16
$\boxed{28 = x}$

Articles sold @ same s.p

(22) A shopkeeper sales 2 T.V sets for same price he gains 20% on one set and loses 20% on another set. what is net loss% & gain% in the whole transaction.

(a) 4% − (b) 6% + (c) 5% − (d) 3% − (e) NOT

$\frac{1}{5}$ ———— (5 : 6) × 2
$\frac{1}{5}$ (5 : 4) × 3 because same s.p

10 : 12
15 : 12
25 : 24

$-\frac{1}{25} \times 100$
= −4%

(23) A person sold 2 article for same price. He gains 20% on 1 article and loss 25% on another article. What is net loss% or profit % in whole transaction?

(a) 15% + (b) 10% − (c) $7\frac{1}{13}$% loss − (d) 12.5% − (e) 5% −

+ 20% = $\frac{1}{5}$ (5 : 6) × 1   5 : 6
− 25% = $\frac{1}{4}$ (4 : 3) × 2   8 : 6
13 : 12

$-\frac{1}{13} \times 100$
$\boxed{= -7\frac{9}{13}\%}$

(24) A person sold 3 articles @ same s.p he got 20% profit, 10% loss and 12.5% profit respectively, on these article. If he got overall ₹ 900 profit, then find the total cp of all the three articles together

(a) ₹ 15200 (b) ₹ 15500 (c) ₹ 15300 (d) ₹15800 (e) NOT

cp	:	sp				
5	:	6) 3	→	15	:	18
10	:	9) ×2	→	20	:	18
8	:	9) ×2	→	16	:	18
				51	:	54

3 gain → 54 17
sp → $\frac{54 \times 3}{8}$

(25) A man sells two articles for same price one on 20% gain another on 25% gain. If one article cost price is 10,000 then what will be the second cp

(a) ₹ 12,000 (b) ₹9600 (c) ₹8000 (d) ₹7200 (e) NOT

cp	:	sp				
5	:	6) ×5	=	25	:	30
4	:	5 ×6		24	:	30

25p → 10,000
24p → $\boxed{9600}$

(26) A person sold 2 articles @ ₹ 4600 each. At the first article he get 15% profit. But he sold the second articles in such a way so that overall he would get neither profit

nor loan. Then find the difference b/w the c.p of both articles
(a) ₹ 6900 (b) ₹ 300 (c) ₹ 1800 (d) ₹ 1200 (e) NOT.

$15\% = \frac{15}{100} = \frac{3}{20}$

	C.P		S.P
1st	20	:	23
2nd	26	:	23 → same s.p

neither profit nor loss so 1st profit hence 2nd 3 part loss

23p → 4600

diff c.p = 26 − 20 = 6p → $\frac{4600 \times 6}{23} = \boxed{₹1200}$

Condition Based problems :—

(27) A shopkeeper sold an article at 30% loss. If he would have bought it Rs. 100 more and sold it ₹ 100 more then loss would be only 20%. Then find the initial cp of an article
(a) ₹ 225 (b) ₹ 230 (c) ₹ 180 (d) ₹ 200 (e) NOT

c.p	s.p
10x + 100	7x + 100

$\frac{30}{100} = \frac{3}{10}$

$(10x+100) \times \frac{4}{5} = 7x + 100$ → 20% loss

$40x + 400 = 35x + 500$

$5x = 100 \rightarrow \boxed{x = 20}$

$c.p = 10x = 10 \times 20 = 200$

(28) A shopkeeper sold his goods @ 20% profit. If he had bought it @ 10% more and sold it for ₹ 27 more. He would have gained 25%. Find c.p of goods
(a) ₹ 160 (b) ₹ 180 (c) ₹ 200 (d) ₹ 240 (e) NOT

c.p	s.p
10x	12x
↓	↓
11x	12x + 27

$\boxed{100} \rightarrow \boxed{110}$
120

$11x \times \frac{5}{4} = 12x + 27$

$55x = 48x + 27 \times 4$

$7x = 27 \times 4$

$\boxed{x = 16}$

$cp = 10x = 10 \times 16 = 160$

(29) Ram sold a watch @ 10% profit. If he bought it @ 10% less and sold it for Rs. 15 more. He would have gained 25%. Find the cp of watch
(a) ₹ 650 (b) ₹ 600 (c) ₹ 120 (d) ₹ 140 (e) NOT

C.P	S.P
10x	11x
↓	↓
9x	11x + 15

$9x \times \frac{5}{4}$ profit

$45x = 44x + 60$

$x = 60$ $10x \rightarrow 10 \times 60 = 600$

(30) A shopkeeper sold an article @ 37.5% profit. If both s.p and c.p decreased by ₹ 40. then you profit would be 2/7th of the s.p. Then find initial cp of the article
(a) ₹ 650 (b) ₹ 675 (c) ₹ 640 (d) ₹ 660 (e) NOT

$37.5\% = \frac{3}{8}$

C.P	=	S.P
8x		11x
↓		↓
8x − 40		11x − 40

profit = $\frac{2}{7}$ s.p

$s.p - c.p = 11x - 40 - (8x - 40) = \frac{2}{7}(11x - 40)$

$7 \times 3x = 22x - 80$

$\boxed{x = 80}$

$cp = 8x = 8 \times 80 = 640$

(30) If s.p of an article is decreased by ₹ 540 but cp is decreased by 20%. Then loss % change from 10% to 20%. Find initial c.p of the price article

ⓐ ₹1800 ⓑ ₹2000 ⓒ ₹2100 ⓓ ₹2200

Cp — Sp

10x — 9x − 520

$8x \times \frac{84}{5} = 9x - 520$

$3x = 45x - 520 \times 5$

$13x = 520 \times 5$

$\boxed{x = 200}$ $10 \times x = 200$

Discount based problems:—

(31) A shopkeeper gave 10% discount on the MP of article thus earning a profit of ₹400. Had he sold the article @ 20% discount, he would have earned 330 as profit. What was the MP of article?

ⓐ 600 ⓑ 900 ⓒ 700 ⓓ 1100 ⓔ 800

MP = 100% → Dis − 10% ⟶ S.p = 90% → 400

Dis = 20% ⟶ S.p = 80% → 300

10% → 70

100% → 700 on MP

(32) MP of article is ₹1600 more than the CP. When the same article is sold at a discount of ₹500 then the % profit earned is 25%. For earning profit of 30%, the article should be sold @ what price?

ⓐ 5740 ⓑ 5720 ⓒ 5620 ⓓ 5730 ⓔ NOT

MP = Cp + 1600 − 500] Cp + 1100

profit = 25% = 1100

C.p → 100% = $\frac{100 \times 1100}{25}$

$Cp = \frac{4400 \times 130}{100}$

profit 30% → = 5720

(33) I asked the shopkeeper the price of a wrist watch. I found that I had just the required sum of money. When the shopkeeper allowed me a discount of 25%, I could bought another watch worth ₹940 for my younger sister. What is the price which I have had paid on my own watch?

ⓐ 2200 ⓑ 1800 ⓒ 2820 ⓓ 3760

Dis → 25% → 940

MP → 100% →

75% → $\frac{940 \times 100}{25}$

$\boxed{\to 2820}$

(34) If MP 20% more than CP and a shopkeeper allows discount of 10% find his profit loss%

ⓐ 15% ⓑ 12.5% ⓒ 20% ⓓ 10% ⓔ Not

Basic :—

CP		MP		SP
100	$\xrightarrow{+20\%}$	120	$\xrightarrow[(-12)]{-10\%}$	108

+8%

Advance (AB method)

$+20 - 10 - \frac{200}{100} = 8\%$

Simple Interest

p = principle
t → time
r → rate

$S.I = \frac{P \times t \times r}{100}$ ✗ $S.I = \frac{p \times t \times r}{100}$

Logic: - on $p = 12,000$ $r = 8\%$ $t = 4$ yrs $S.I = ?$ $A = ?$

$S.I = t \times r$

$S.I = 8 \times 4 = 32\%$ of 12,000

Amount $= 12,000 \times \frac{132}{100} = 15840$

② $p = ?$, $r = 8\%$ $t = 4$ yrs Am = 15840
$= \boxed{100\%}$

$p = 100\%$ $S.I = r \times t = 32\%$

Amount $= 100\% + 32\% = 132\% \longrightarrow 15840$

$100\% \longrightarrow \frac{15840 \times 100}{132}$

$= 12,000$

Problems: →

① A sum fetched a total S.I of ₹3200 at the rate of 6.25% per annum in 4 yrs. What is the sum in (Rs)?

(a) 13,800 (b) 11800 (c) 12800 (d) 14800 (e) NOT

$S.I = r \times t = 4 \times 6.25 = 25\% \longrightarrow 3200$

$100\% \longrightarrow \frac{3200 \times 100}{25}$

$\boxed{p = 12,800}$

② Lakshman borrowed certain a sum of money at S.I at the rate of 5% p.a for 1st 3 yrs 9% p.a next 5 years and 15% p.a beyond 8 yrs. If the total interest paid by him at the end of 12 yrs is Rs 4800. how much money did he borrow? in Rs

(a) 4000 (b) 2000 (c) 4500 (d) 5600 (e) 4840

1st 3 yrs → $5\% \times 3 = 15\%$

next 5 yrs → $9\% \times 5 = 45\%$

beyond 8 yrs → $15\% \times 4 = 60\%$

$S.I = 120\% \longrightarrow 4800$

$100\% \longrightarrow \frac{4800 \times 100}{120}$

$\boxed{p = 4,000}$

③ Two equal sum of money were invested, one at 4% and other @ $4\frac{1}{2}\%$ at the end of 7 yrs, the S.I received from the later exceed that received from the former by ₹ 31.50. Each sum was in Rs

(a) ~~3000~~ 1000 (b) 500 (c) 750 (d) 900 (e) NOT

Basic → $4\% \times 7 = 28$

$4\frac{1}{2} \times 7 = 31.50$

$3.5\% \longrightarrow 31.50$

$100 \longrightarrow \frac{31.50}{3.5} \times 100 = 900$

Logic → $\frac{1}{2}\% \times 7 = 3.5\%$

④ In how many yrs S.I obtained on ₹ 8000 @ 3% per annum will be equal S.I on ₹ 6000 @ 4% for 5 yrs?

(a) 4 years (b) 5 years (c) 6 years (d) 8 years (e) NOT

$S.I = r \times t = 4 \times 5 = 20\%$

$\longrightarrow 6000 \times \frac{20}{100} = \boxed{1200} \rightarrow S.I$

$S.I = \frac{ptr}{100} = \frac{8000 \times t \times 3}{100} = 1200$

$\boxed{t = 5 \text{ yrs}}$

⑤ A person lent sum of money @ 5% p.a S.I and 15 yrs the interest amounted to ₹ 250 less than sum lent what was the sum lent (in Rs)

(a) 1000 (b) 1500 (c) 2400 (d) 3000 (e) NOT

$p = 100\%$ $S.I = 5\% \times 15 = 75\%$

$25\% \longrightarrow 250$

$100 \rightarrow \boxed{1000}$

⑥ A person lent certain sum of money @ 9.5% p.a S.I for 40 yrs the interest amounted to ₹ 22,400 more than the sum lent. What was the sum lent? in Rs

ⓐ 8100 ⓑ 8500 ⓒ 8000 ⓓ 8700 ⓔ NOT

$P = 100\%$ $9.5\% \times 40 = 380\% = S.I$

$280\% \rightarrow 22,400$

$100 \rightarrow \frac{22,400 \times 100}{280} = \underline{8000}$

⑦ If a certain sum of money become three itself in 5 yrs 4 months @ S.I, then what will be yearly rate of interest? (%)

ⓐ 18.75% ⓑ 37.5% ⓒ 37.5% ⓓ 42.25% ⓔ NOT

$P = 100\% \xrightarrow{\times 3} A = 300\%$

$200\% = S.I$

$t = 5\frac{4}{12} = 5\frac{1}{3} = \frac{16}{3}$ yrs

$t \rightarrow$ Int

$\frac{16}{3}$ yrs $\rightarrow 200\%$

1 yr $\rightarrow \frac{200 \times 1}{16} \times 3$

$= 37.5$

⑧ At a rate of S.I the interest on a sum of money in 10 yrs will be $\frac{3}{5}$th part of the amount. Then rate of S.I per annum is (in %)

ⓐ 30% ⓑ 15% ⓒ 10% ⓓ 7.5% ⓔ NOT

$P = 100\%$

$\text{Interest} = \frac{3}{5} \times A$

$\frac{I}{A} = \frac{3}{5}$

$P = 2$ 2 part principle $\longrightarrow$ 5 part amount

$100\% \longrightarrow \frac{5 \times 100}{2}$

$= 250\%$

$150\% \rightarrow S.I$

⑨ A certain sum money triples in 5 yrs @ S.I. In how many yrs it will be five times?

ⓐ 5 yrs ⓑ 8 yrs ⓒ 10 yrs ⓓ 15 yrs ⓔ NOT

$P = 100\% \longrightarrow 300\%$ 2nd cond. $P = 100 \longrightarrow 500$

$400\% \rightarrow ?$

$200\% \longrightarrow 5$ yrs

$400\% \longrightarrow ? = \frac{5 \times 400}{200} = 10$ yrs

⑩ Sumit lent some money to Mohit @ 5% annum S.I. Mohit lent the entire amount to Brijesh on the same rate @ day @ $8\frac{1}{2}\%$ per annum. In this transaction after a yr Mohit earned profit of ₹ 350. Find the sum of money lent by Sumit to Mohit in Rs

ⓐ 10,000 ⓑ 9000 ⓒ 10,200 ⓓ 18,600 ⓔ NOT

$S \xrightarrow{5} M \xrightarrow{8\frac{1}{2}\%} B$

$3.5\% \rightarrow 350$

$100\% \rightarrow \frac{350 \times 100}{3.5} = \boxed{10,000 = S}$

⑪ A certain sum of money amounts to ₹ 919 in 2 yrs and ₹ 969 in 3.5 yrs @ S.I. What is the ratio of interest (%)?

ⓐ 4 ⓑ 5 ⓒ 6 ⓓ 8 ⓔ NOT

sum $\xrightarrow{2 yrs}$ 919

sum $\xrightarrow{3.5 yrs}$ 969

1.5 yrs $\longrightarrow$ 51

1 yr $= \frac{51}{1.5} = 34$ Rs

S.I in 2 yrs $\rightarrow 2 \times 34 \rightarrow 68 \rightarrow S.I$ $A = 919$

$919 - 68 = 850 = P$

$= \frac{34}{850} \times 100$

$\boxed{\% = 4\%}$

(13) S.I on a sum of money will be ~~25~~ 225 after 3 yrs. In the next 5 yrs amount becomes 3 times of principle. What will be the amount in 8 yrs? Ans

(a) 1275 (b) 900 (c) 1735 (d) 985 (e) Not

S.I $\longrightarrow$ 3 yrs $\longrightarrow$ 225

3 yr $\longrightarrow$ 75

$\boxed{1} \xrightarrow{3\,yrs} \boxed{} \xrightarrow{5\,yrs} \boxed{3}$

5 yrs $\longrightarrow$ 5×75

= 600 $\longrightarrow$ 2p [3 times − 1 time]

$\boxed{900}$ $\longleftarrow$ 3p

(14) At S.I ₹800 becomes 920 in 3 yrs. If rate of interest is increased numerically by 3% then total amount will become Ans

(a) 972 (b) 982 (c) 992 (d) 1002 (e) Not

$800 \xrightarrow{3yr} 920$

3 yrs $\rightarrow$ 3%

3 yrs = 9%

$\frac{800 \times 9}{100} = 72$

Amount will be 920 + 72

= 992

(15) A sum of Rs. 400 becomes Rs 448 @ S.I in 2 years. In how many yrs will be the sum of ₹550 amount to ₹682 at the same rate?

(a) 2 yrs (b) 3 yrs (c) 3.5 yrs (d) 4 yrs (e) Not

48/2 = 24

$400 \xrightarrow{2yrs} 448$

rate = $\frac{24}{400} \times 100 = \boxed{6\%}$

$550 \longrightarrow 682$

$\frac{132}{550} \times 100$ $\frac{132}{550} \times 100 = \boxed{24\%}$

4 yrs

(15) ~~2~~ 2200 rupees become ~~Rs~~ 2464 in 1.5 yrs at S.I and the principle amount that will be become 9240 rupee in 4 years at the same rate of interest.

(a) 9000 (b) 8000 (c) 7000 (d) 5000 (e) Not

$2200 \xrightarrow[(1.5\,yrs)]{} 2464 \rightarrow \frac{264}{1.5} = 176 = \frac{176}{2200} \times 100 = 8\%$

$P \xrightarrow[(4\,yrs)]{} 9240$

×8%

$\boxed{100\%}$ 32% $\boxed{132\%}$

132% $\longrightarrow$ 9240

100% $\longrightarrow$ $\frac{9240 \times 100}{132}$

$\boxed{P = 7000}$

(6) If a sum of money at S.I becomes 7/5 times in 4 yrs. Then it will become 17/10 times in (yrs)

(a) 13 (b) 7 (c) 14 (d) 3.5 (e) Not

principle should be same

(5 $\longrightarrow$ 7) ×2

2p = S.I

2p $\longrightarrow$ 4 yrs

7p $\longrightarrow$ $\frac{4 \times 7}{2}$ = 14 yrs.

1× (10 $\longrightarrow$ 17)

7p → S.I

4

10 $\longrightarrow$ 14 4p $\rightarrow$ 4 yrs

10 $\longrightarrow$ 17 7p $\rightarrow$ $\frac{4 \times 7}{4}$ = $\boxed{7\,yrs}$

(7) A certain sum was invested on S.I. The amount to which it had grown in 6 yrs was $\frac{7}{6}$ times the amount to which it had grown in 4 yrs. The percentage rate of interest was:

(a) 12.5% (b) 13.5% (c) 14.5% (d) 21.5% (e) Not

$A6 = \frac{7}{6} A4$

$\frac{A6}{A4} = \frac{7}{6}$)2

2 yrs $\longrightarrow$ 1p

in 1p $\longrightarrow$ $\frac{1}{2}$ p

$\boxed{5}$ $A6 \xrightarrow{6yrs}$ ~~14~~ 7

$\boxed{6}$ $A4 \xrightarrow{4yrs}$ 6

2 yrs → 1p

4 yrs $\longrightarrow$ 2p 4p $\xrightarrow{(2P)}$ 6

$\frac{1}{2 \times 4} \times 100$

= 12.5%

(13) When a person invests some money under S.I then at the end of some years the amount become 13 times of the principle and the numerical value of rate of interest per annum is thrice of the time. At the end of 10 yrs, the amount will become how many times of the principal at the same rate of interest?

(a) 6 times (b) 8 times (c) 9 times (d) 7 times (e) NOT

$S = 100 \longrightarrow 1300 = A$ $t = x$

$S.I = 1200$ $r = 3x\%$

$S.I = r \times t = 3x^2$

$1200 = 3x^2$

$\boxed{x = 20}$ $r = 3x$ $\boxed{r = 60\%}$

2nd case $p = 100 \xrightarrow[60 \times 10\% = 600\%]{10 yrs} 700$

7 times of principle

(14) A money lender claims to lend money at the rate of 12.5% per annum S.I. However, he takes the interest in advance when he lends a sum for one year. At what interest rate does he lend the money actually?

(a) 12.14% (b) 28.14% (c) 14.29% (d) 19% (e) NOT

$12.5\% \rightarrow \frac{1}{8}$

customer point of view

$p = 7$

$I = 1$ $r = \frac{1}{7} \times 100 = 14.29\%$

(20) The rate of S.I p.a. of bank being increased from 8.5% to 15% and time duration is reduced from 3.5 yrs to 13/5 years. The total S.I was increased by 1480. Find the initial S.I in Rs

(a) 6047 (b) 4760 (c) 7460 (d) 7640 (e) NOT

1480

(60-30.25)

$8.5\% \times 3.5$ yrs | 15% $\frac{13}{5}$ yrs

$S.I = r \times t$

$= 29.75$ | $= 39\%$

$9.25\% \longrightarrow 1480$

$\rightarrow 29.75 \longrightarrow \frac{1480 \times 29.75}{9.25} = \underline{\underline{4760}}$

(21). A sum of 12,000 invested partly at 8% annum and the remaining @ 11% per annum simple interest. If the total interest rate at the end of 2.6 yrs is ₹ 3120. How much money was invested at 11% per annum?

(a) 2000 (b) 3250 (c) 8000 (d) 3000 (e) NOT

Basic

12000

(12000 − x) (x)

$(12000 - x) \times \frac{8}{100} + \frac{x \times 11}{100} = 1200$

$960 + \frac{3x}{100} = 1200$

$\frac{3x}{100} = 240$

$\boxed{x = 8000}$

2.6 yrs → 3120

1 yrs = 1200

rate of $= \frac{1200}{12000} \times 100$

overall rate of $= 10\%$

8% 11%

$\boxed{10\%}$

1 : 2

3p → 12,000

1p → 4000

hence 2p = 8000

(22) Ravi gave 1200 on loan. Some amount he gave @ 4% p.a and remaining @ 5% p.a. S.I. After 2 yrs, he got Rs. 110 as interest. The amount given @ 4% and 5% respectively are

(a) 500, 700 (b) 400, 800 (c) 800, 400 (d) 1100, 100 (e) NOT

2 yrs ⟶ 110 Rs

1 yr ⟶ 55 Rs ⟶ S.I rate% $= \frac{55}{1200} \times 100$

$= \frac{55}{12}\%$

~~40~~ 12 × (4%)   (5%) × 12

$(5\frac{5}{12}\%) \times 12$

48%   60%

55%

5 : 7 $= 12p = 1200$

500 : 700 $\boxed{p = 100}$

(23) Arun lends Rs. 20,000 to two of his friends for 2 years. 12,000 to the first @ 8% p.a. S.I. Arun wants to make profit of 10% on the whole. The S.I rate at which he should lend the remaining sum of money to the second friend is

(a) 8% (b) 16% (c) 12% (d) 30% (e) NOT

(12,000) 1st   2nd (8,000)

$\boxed{8\%}$ $\boxed{x\%}$

whole $\boxed{10\%}$

$\boxed{x - 10\%}$: $\boxed{2\%}$

3 : 2   2 → 3

$\frac{x-10}{2} = \frac{3}{2}$

$2x - 20 = 6$

$2x = 26$

$\boxed{x = 13\%}$

(24) A person invested ~~1/4 the of the~~ one-fourth of the sum of Rs. 20,000 at certain rate of S.I and rest at 3% p.a. higher rate. If the total received for 2 yrs is ₹ 4100, what is the rate at which the second sum was invested?

(a) 8% (b) 11% (c) 6% (d) 2% (e) NOT

1st   2nd

$\boxed{x\%}$ $\boxed{x + 3\%}$

overall

$\boxed{10.25\%}$

$\boxed{x - 7.25}$ $\boxed{~~8~~ - 10.25\% - x}$

1 : 3

2 yrs = 4100

I = 2050   10.25

$rate\% = \frac{2050}{20,000} \times 100$

$\% = 10.25\%$

$\frac{x - 7.25}{x - 10.25} = \frac{1}{3}$

$3x - 21.75 = ~~3~~ - 10.25\% - x$

$4x = 32$

$\boxed{x = 8}$

2nd sum

$x + 3\% = 8 + 3\%$

$= 11\%$

COMPOUND INTEREST :→

SI → Interest → principal

CI → Interest → Amount

Amount = principle + Interest

* S.I for every year remains same
* C.I for every year to Increase
* S.I & C.I for the first year remains same.

Formula :→ (Basic)

$CA = P\left(1+\frac{r}{100}\right)^t$ ⇒ $CI = CA - P$

$= P\left(1+\frac{r}{100}\right)^t - P$

$= P\left[\left(1+\frac{r}{100}\right)^t - 1\right]$

Ex- $P = 10{,}000$, $T = 3$ yrs, $R = 10\%$

A = ? C.I = ?

M-1

$CA = 10{,}000\left(1+\frac{10}{100}\right)^3$

$= 10{,}000 \times \frac{110}{100} \times \frac{110}{100} \times \frac{110}{100}$

1st yr, 2nd yr, 3rd yr

$= 13310$

$C.I = CA - P$

$= 13310 - 10{,}000$

$= \underline{3310}$

M-2 (Successive)%

$\left(a+b+\frac{a \times b}{100}\right)\%$

$= 10+10+\frac{100}{100} = \underline{21}\%$

$+21+10+\frac{210}{100} = \underline{33.1}\%$

$C.I = 10{,}000 \times \frac{33.1}{100} = 3310$

$C.A = \cancel{13310} \cdot 10{,}000 + 3310 = \underline{13310}$

M-3 (Tree method) (learning)

$P = 10{,}000$ $r = 10\% = \frac{1}{10}$

1st, 2nd, 3rd

1,000 1,000 1,000 → S.I = 3,000

10% → 100, 100, 100, 10

Total C.I = $\underline{3310}$

Base (method)

$r = 10\% = \frac{1}{10}$ for 3 yrs

P → 10 : 11

10 : 11

10 : 11

P ← 1000 : 13,310 ← A

331 → C.I part

1000 P → 10,000

331 → $\frac{10{,}000 \times 331}{1000}$

C.I = Rs $\underline{3310}$

(1) The C.I on ₹ 2000 in 2 yrs if the rate of interest is 4% p.a and for 1st yr and 3% p.a for 2nd yr, will

P → 2000 t = 2 yr r = 4% r = 3%. C.I = ?

M → 1 $2000 \times \frac{104}{100} \times \frac{103}{100} = 1071282$

2142.4 → C.A

$I = 2142.4 - 2000$

$= \underline{142.4}$

M → 2 $+4+3+\frac{12}{100} = 7.12\%$ $C.I = 2000 \times \frac{7.12}{100}$

$= \underline{142.4}$

M → 4 $r_1 = 4\% = \frac{4}{100} = \frac{1}{25}$ $r_2 = 3\% = \frac{3}{100}$

25 : 26

100 : 103

2500 : 2678

C.I = 178 P

2500 → 2,000

1 P → $\frac{2{,}000}{2500} = \frac{4}{5}$

178 → $\frac{4}{5} \times 178 = 142.84$

② A certain sum amounts to ₹ 5832 in 2 yrs @ 8% p.a compound interest, the sum is

$p \to ?$ $A \to 5832$ $t = 2yr$ $r = 8\% = \frac{2}{25}$

1st yr 25 : 27
2nd yr 25 : 27
625 : 729

$729 \to 5832$
$625 \to \frac{5832 \times 625}{729}$

$p = 5,000$

③ The C.I on ₹ 8000 @ 15% p.a for 2 yrs 4 months, compounded anually is

$p = 8000$, $t = 2$ yrs 4 months $C.I = ?$ $r = 15\% = \frac{15}{100} = \frac{3}{20}$

1st → 20 : 23
2nd → 20 : 23
4 months → 20 : 21
8000 : 11109

12 months → 3p
4 months → $\frac{3 \times 4}{12} = 1p$

3109 → 3109 ₹

$8000 \times \frac{115}{100} \times \frac{115}{100} \times \frac{105}{100}$

12 m → 12 m
4 m → $\frac{12 \times 4}{12}$

12 m → 15%
4 m → $\frac{15 \times 4}{12} = 5\%$

If (2 yrs 73 days)
365 → 15%
73 → $\frac{15 \times 73}{365}$

④ The certain sum, invested @ 4% p.a C.I, compounded half yearly, amounts to ₹ 7803 at the end of one yr. The sum is

$r = 4\%$ half-yearly, $CA = 7803$ $t = 1yr$ $P = ?$

If Half-yearly ⇒ Rate/2
Time ⇒ Time × 2

New rate $= 2\% = \frac{2}{100} = \frac{1}{50}$ New time $= 1 \times 2 = 2$ yrs

1st yr → 50 : 51
2nd yr → 50 : 51
2500 : 2601

2601 → 7803
2500 → $\frac{7803 \times 2500}{2601}$

$\boxed{p = 7500}$

⑤ The C.I on ₹ 30,000 @ 7% p.a C.I for certain time is ₹ 4347, The time is

$p = 30,000$, $r = 7\%$ $CI = 4347$ $t = ?$

$r = \frac{7}{100}$

Base :→ 100 : 107 → 100 : 107
(100 : 107) → 2nd yr
on n yr → 30,000 : 34347 → 30,000 : 34347
+ 4347
1000 : 11449

Total 2 yrs

⑥ The sum of ₹ 8000 amount to 8820 in 2 yrs if the interest is calculated annually. The rate of C.I is

$p = 8000$, $CA = 8820/-$ $t = 2$ yrs $r = ?$

20 : 21
20 : 21

Base :→
2nd yr :→ 8000 : 8820 → $\sqrt{400}$: $\sqrt{441}$
$\sqrt{400}$ $\sqrt{441}$ →

20 : 21

$(r\%) = \frac{1 \times 100}{20} = \boxed{5\%}$

⑦ At what rate p.a ₹ 8000 at 15% p.a annum for 2 yrs, 4 months, compound anually

(7) At what rate p.a ₹ 32,000 yield C.I of ₹ 5044 in 9 months interest being CA compounded quarterly

$p = 32{,}000$ $r = ?\%$ → quarterly

$C.I = 5044$ $t = 9$ months

For quarterly :- Rate → Rate/4 Time → Time × 4

New time = $9 \times 4 = \frac{36 \text{ months}}{12} = 3$ yrs

New rate :→ $\frac{\text{original rate}}{4}$

3rd yr :→ 32,000 : 37044

for 3 yrs $= \sqrt[3]{8000}$: $\sqrt[3]{9261}$

= 20 : 21

rate% $= \frac{1}{20} \times 100 = 5\%$

~~New rate~~ = original rate = $5 \times 4 = 20\%$

(8) A sum of money on C.I amounts to ₹ 10647 in 3 yrs and ₹ 9680 in 2 yrs. The rate of interest per annum is ?

3 yrs → 10,647

2 yrs → 9680

$p = ?$ $r = ?$ CA

$P(1+\frac{r}{100})^{3} = 10{,}647$

$P(1+\frac{r}{100})^{2} = 9680$

$\frac{r}{100} = \frac{10648}{9680} - 1$

$\frac{r}{100} = \frac{968}{9680}$ ⇒ $\boxed{r = 10\%}$

(or)

2 yrs	3 yrs
9680	10648

$\frac{968}{9680} \times 100 = 10\%$ → C.I calculated on previous year only (or)

3−2 → 1 yr diff

(9) A sum of money @ C.I triples itself in 8 yrs. In how many yrs it will be amount to 243 times itself?

sum → 3 times (8 yrs) → 243 times (time = ?)

x → $3x$ → $9x$ → $27x$ → $81x$ → $273x$

8 yrs + 8 yrs + 8 yrs + 8 yrs + 8 yrs

= 40 yrs.

logic = 3 → 243

3^5 → 243 1 power → 8 yrs

5th power → $8 \times 5 = 40\%$

(10) If a certain sum of money becomes equal to 64 times of itself in 27 years. In how many much time it will become 512 times itself

64 times → 27 years

512 times → ? years

8×8 times → 27 yrs

13.5 + 13.5

$8 \times 8 \times 8$ times → $\boxed{40.5 \text{ years}}$

13.5 + 13.5 + 13.5

(11) If a certain sum of money amounts to ₹ 4900 in 4 yrs and ₹ 7350 in 8 yrs compounded anually. Then in 12 yrs it will become how much?

$x \xrightarrow{4\,yrs} 4900 \xrightarrow{4\,yrs} 7350 \xrightarrow{4\,yrs} ?$

12 yrs

$\text{Times} = \frac{7350}{4900} = \frac{3}{2}$ every 4 years $\rightarrow \frac{3}{2}$ times

$\frac{\overset{3675}{\cancel{7350}} \times 3}{\cancel{2}} = 11025$

To find principle x

$= x \times \frac{3}{2} = 4900 \Rightarrow x = 4900 \times \frac{2}{3}$

(12) ₹ 2700 becomes ₹ 4800 at a certain rate of interest compounded anually in 3 yrs. Find the amount at the end of 4.5 year

$2700 \xrightarrow{3\,yrs} 4800$

$\underbrace{2700 \quad \bigcirc}_{1.5\,yr} \underbrace{\quad 4800}_{1.5\,yr} \underbrace{\quad ?}_{1.5}$ sum

$1.5\,yr \longrightarrow x$ times

$2700 \times x \times x = 4800$

$2700 \times x^2 = 4800$

$x^2 = \frac{\cancel{48}\,16}{\cancel{27}\,9}$

$\boxed{x = \frac{4}{3}}$ here $= \overset{16}{\cancel{4800}} \times \frac{4}{3} = 6400$

(13) ₹ 3200 becomes ₹ 4000 at a certain rate of interest compounded anually in 1.5 year. Then in how many years ₹ 12800 will amounts to 25,000 at the same rate of interest

$3200 \xrightarrow{1.5\,years} 4000$ $\qquad$ $12800 \xrightarrow{time?} 25,000$

$\text{No. of times} = \frac{4000}{3200} = \frac{5}{4}$ $\qquad$ $\text{No. of times} = \frac{25,000}{12,800} = \frac{125}{64}$

$\frac{5}{4}$ time $\longrightarrow$ 1.5 year

$\frac{125}{4}$ time $\rightarrow$ $\frac{1.5 \times \frac{\overset{25}{\cancel{125}}}{\cancel{64}\,16}}{\frac{5}{4}} = 1.5 \times \frac{25}{16}$

(or)

$\left(\frac{5}{4} \times \frac{5}{4} \times \frac{5}{4}\right) = \frac{125}{64}$

$1.5 + 1.5 + 1.5 = 4.5$ yrs

(S.I – C.I Difference (combined))

① The simple interest on a certain sum of money for 2 yrs @ 6% p.a is Rs. 300. The compound interest at the same rate for the same time will be

sum → ? $t = 6$ yrs $r = 6\%$ S.I = 300

Basic = C.I = ?

$S.I = \frac{P \times t \times r}{100}$

$C.I = \left[P\left(1+\frac{r}{100}\right)^t - 1\right]$

or Advanced (successive % & C.I → %)

$S.I = 6 \times 2 = 12\%$

$C.I = +6+6+\frac{6\times6}{100} = 12.36$

$C.I - S.I = 12.36 - 12 = 0.36$

~~C.I = 300~~

$12\% \rightarrow 300$

$12.36 \rightarrow \frac{300 \times 12.36}{12.00} = 300.$

$= 309$

② On a certain sum of money compound interest earned after 2 yrs is ₹ 3816 and S.I for after 3 yrs is ₹ 5400. Find r = ? if the rate of interest is same

For 2 years

		1st	2nd	3rd yr
C.I	S.I →	1800	1800	1800
	C.I →	1800	2016	

1st yr same

216 → C.I

$\frac{216}{1800} \times 100 = 12\%$

or $\boxed{\frac{S.I}{C.I} = \frac{200}{200+r}}$ for 2 yrs

$\frac{3600}{3816} = \frac{200}{200+r}$

$\boxed{r = 12\%}$

③ The S.I on certain sum of money for 2 yrs @ 6% p.a is ₹ 300. The C.I at the same rate for the same time is

$\frac{S.I}{C.I} = \frac{200}{200+r}$ $\frac{300}{C.I} = \frac{200}{206}$

C.I = 648 ~~306~~ 309

④ What is the difference b/w maturity values, if ₹ 12,500 is invested in 2 years @ 20% p.a S.I and C.I?

$P = 12{,}500$ $t = 2$ $r = 10\%$

Formula = Basic $S.I = \frac{Ptr}{100}$ $C.I = \left\{P\left[\left(1+\frac{r}{100}\right)^t - 1\right]\right\}$.

Formula :– $diff = P\left(\frac{r}{100}\right)^2$ For 2 yrs only

$= 12{,}500\left(\frac{20}{100}\right)^2$

$= 12{,}500\left(\frac{1}{25}\right)$

$= \underline{500}$

or successive

S.I = for 2 yrs = 2×20 = 40%

C.I = 20+20+$\frac{400}{100}$ = 44%

4% of 12,500

diff = 500

⑤ The difference b/w C.I and S.I on a sum for 2 yrs @ 20% per annum is ₹ 200. If the interest is compounded half yearly, then what is the difference (ines) b/w C.I and S.I for 1 year?

C.I – S.I = 200 $r = 20\%$ $t = 2$ yrs

$diff = P\left(\frac{r}{100}\right)^2 = 200 = P\left(\frac{20}{100}\right)^2$

$200 = P\left(\frac{1}{25}\right)$

$P = 5000$

Half yearly = rate = $\frac{1}{2}$ rate = 10% time = 2t = 2 yr

$diff = 5000\left(\frac{10}{100}\right)^2$

$= 5000\left(\frac{1}{100}\right)$

$= \underline{50}$

(6) What is the difference (in Rs). C.I anually and S.I for 3 years on a principle Rs.3000 at the rate of 20%

Diff CI - SI = ?

$p = 3001 \quad t = 2yrs \quad r = 20\%$

successive (of rate)

$diff = \frac{pr^2(300+r)}{100^3}$

$= \frac{3000 \times 20 \times 20 \times 320}{1000000}$

$= 384$

S.I = 20% × 3 = 60%

$C.I = 20 + 20 + \frac{400}{100} = 44\%$

$+ 44 + 20 + \frac{44 \times 20}{100} = 72.81\%$

12.81% of 300

$\boxed{diff = 384}$

(7) The difference of C.I and S.I for 3yrs and for 2yrs are in ratio 23:7 respectively. What is the rate of interest p.a?

diff CI & S.I for 3 years / " " " for 2 years $= \frac{23}{7}$

$\frac{\frac{pr^3(300+r)}{100^3}}{P(\frac{r}{100})^2} = \frac{pr^2(300+r)}{100^3} \times \frac{100^2}{pr^2}$

$= \frac{300+r}{100} =$

$\frac{300+r}{100} = \frac{23}{7}$

$2100 + 7r = 2300$

$(r = \frac{200}{7})$

$= 28\frac{4}{7}\%$

(8) A sum of ₹ 7,500 amount to ₹ 8748 after 2yrs at a certain C.I @ rate per annum. What will be the S.I on the same sum for 5yrs at double the earlier interest rates rates

(a) 8000 (b) 6000 (c) 4000 (d) 2,700

7500 ⟶ 8746

50² : 54²

$\sqrt{2500} : \sqrt{2916}$

50 : 54

$\frac{4}{50} \times 100 = 8\%$

S.I = ? r = 16% sum = 7500 t = 5yrs

S.I = t×r = 16×5 = 80%

$S.I = 80\% \text{ of } \frac{80}{100} \times 7500$

$\boxed{S.I = 6000}$

(9) For an amount, S.I at the rate of interest 12% p.a for 6yrs is ₹ 25920. What will be the C.I on same amount at the rate of interest of 8% p.a compounding anually for 2yrs?

(a) 4263.3 (b) 5563.4 (c) 5888.6 (d) 5990.4

S.I ⟶ 12×6 = 72% ⟶ 25,920

$C.I \longrightarrow 8 + 8 + \frac{16}{100} = 16.64\%$

$\boxed{\begin{array}{l} 72\% \longrightarrow 25,920 \\ 16.64\% \longrightarrow \frac{25920 \times 16.64}{72} \end{array}} = 5990.4$

(10) The S.I on a certain sum for $3\frac{1}{2}$ years @ 10% p.a is ₹ 2940. What will be the C.I on the same sum for $2\frac{1}{2}$ yrs at the same rate when interest compounded yearly (nearest to rupee)?

(a) ₹ 2327 (b) ₹ 2272 (c) ₹ 2227 (d) ₹ 2372

S.I = 10% × 3.5 = ~~35~~ 35% ⟶ 2940

C.I ⟶ t = $2\frac{1}{2}$ years for 2 years = $10+10+\frac{100}{100}$

= 21%

for $\frac{1}{2}$ year 10% ⟶ 1 year

5% ⟵ $\frac{1}{2}$ year

for $2\frac{1}{2}$ years C.I = $21+5+\frac{21\times5}{100}$

= 27.05%

35% ⟶ 2940

27.05% ⟶ $\frac{2940\times27.05}{35}$

(11) Mr. Khandelwal invested a total ~~a~~ sum of ₹16,000 in two schemes D_1 and D_2 for 2 years. D_1 offers C.I at the rate of 10% p.a and D_2 offers S.I at the rate of 12% p.a. He earned a total of ₹ 3504 at the as interest from both the schemes. Find the amount invested him in D_2

Basic:- 16,000

D_1	D_2
C.I	S.I
r=10%	r=12%
p = 16000−x	p=x

$\boxed{\frac{p r t}{100} + p\left[\left(1+\frac{r}{100}\right)^t - 1\right] = 3500}$

or : Allegation

for (2 years)

D_1		D_2
21%		24%

overall

$\boxed{21.9\%}$

2.1 : 0.9

7 : 3

10 ⟶ 16,000

3 ⟶ $\frac{16000\times3}{10}$

$\boxed{D_2 = 4800}$

Time and distance :—

Speed = $\frac{\text{Distance}}{\text{Time}}$ $\boxed{T = \frac{d}{s}}$ ~~D×T×~~ $\boxed{D = T \times S}$

① In covering certain distance, the speed of A and B are in the ratio 3:4 A takes 30 min more than B to reach the destination. The time taken by A to reach the destination

ⓐ 80 min ⓑ 90 min ⓒ 120 min ⓓ 150 min ⓔ 60 min

A : B

S → 3 : 4

T → 4 : 3

1p → 30 min

4p ⇒ 120 min

② A train running 9/13th of its own speed reached a place in 39 hours. How much time could be saved if the train would run its own speed?

ⓐ 20 ⓑ 12 ⓒ 16 ⓓ 25 ⓔ NOT

Current speed = original speed × $\frac{9}{13}$

$\frac{C}{O} = \frac{9}{13}$

C : O

S → 9 : 13

t → 13 : 9

4p saved

13p → 39 hours

4p → $\frac{39 \times 4}{13}$ = 12 min

③ A boy goes to school @ a speed of 5km/h and return to the village at a speed @ 4km/h. If he takes 4 hours and 30 min in all what is the distance between village and school in km

ⓐ 7 ⓑ 10 ⓒ 15 ⓓ 4 ⓔ NOT

→ ←

S → 5 : 4

t → 4 : 5

9p → 4.5 hours

4p → 4.5×4 ...

d = S×t

= 5×2

= 10 km

④ Ravi and Ajay starts ~~immediately~~ simultaneously, from a place A toward B, 60 km apart. Ravi's speed is 4 km/h less than that of Ajay. Ajay after reaching B, turns back and meets Ravi at place 12 km, away from B. Ravi's speed is?

ⓐ 12 km/h ⓑ 8 km/h ⓒ 24 km/h ⓓ 16 km/h

A ← 60 km → B, ← 12 km →

$\boxed{R = A - 4}$

t = constant

	Ajay	Ravi
	60+12	60−12
D =	72	48
D =	3	2
S =	3	2

1p → 4 km/h

2p → 8 km/h

⑤ Distance b/w A and B is 96 km. Two person p and q having speed 50 km/h and 70 km/h starts from A to B simult. q reached B and returned immediately and met p at distance x from B. find the value of x

A ← 96 km → B; ← 5p →, x, ← 7p →

p = 50 km/h

q = 70 km/h

t = constant

	p	q
S =	50	70
S =	5	7
D =	5	7

6p → 96 km

2p → $\frac{96 \times 2}{6}$ = 16

$\boxed{x = 16 \text{ km}}$

⑥ A man covered one third of his journey at a speed of 20 km/h, one fourth of the remaining at the speed of 25 km/h, and the remaining journey at a speed of 30 km/h. Find the avg speed of man.

Time and distance :—

$Speed = \frac{Distance}{Time}$ $\boxed{T = \frac{d}{s}}$ ~~DxT~~ $\boxed{D = T \times S}$

① In covering certain distance, the speed of A and B are in the ratio 3:4 A takes 30 min more than B to reach the destination. The time taken by A to reach the destination

ⓐ 90min ⓑ 90min ⓒ 120min ⓓ 150min ⓔ 60min

A : B
S → 3 : 4
T → 4 : 3
1p → 30min
4p ⇒ 120min

② A train running 9/13th of its own speed reached a place in 39 hours. How much time could be saved if the train would run its own speed!

ⓐ 20 ⓑ 12 ⓒ 16 ⓓ 25 ⓔ NOT

Current speed = original speed × $\frac{9}{13}$

$\frac{C}{O} = \frac{9}{13}$

C : O
S → 9 : 13
t → 13 : 9
4p saved
13p → 39 hours
4p → $\frac{39 \times 4}{13}$ = 12 min

③ A boy goes to school @ a speed of 5km/h and return to the village at a speed @ 4km/h. If he takes 4 hours and 30min in all what is the distance between village and school in km

ⓐ 7 ⓑ 10 ⓒ 15 ⓓ 4 ⓔ Nor

S → 5 : 4
t → 4 : 5
9p → 4.5 hours
4p → $\frac{4.5 \times 4}{9}$ = 2 hours

d = S×t
= 5×2
= 10 km.

④ Ravi and Ajay starts ~~immediately~~ simultaneously from a place A towards B, 60 km apart. Ravi's speed is 4km/h less than that of Ajay. Ajay after reaching B, turns back and meets Ravi at place 12km away from B. Ravi's speed is!

ⓐ 12km/h ⓑ 8km/h ⓒ 24km/h ⓓ 16km/h

A ← 60km → B
← 12km →

$\boxed{R = A-4}$

t = constant

	Ajay	Ravi
D =	60+12 = 72	60−12 = 48
D =	3	: 2
S =	3	: 2

1p → 4km/h
2p → 8km/h

⑤ Distance b/w A and B is 96km. Two person P and Q having speed 50km/h and 70km/h starts from A to B simult. Q reached B and returned immediately and met P at distance x from B. Find the value of x

A ← 96km → B
← 5p → ← 1p →
x ← 1p →

P = 50km/h
Q = 70km/h

t = constant

	P	Q
S =	50	70
S =	5	: 7
D =	5	: 7

6p → 96km
1p → $\frac{96}{6}$ = 16

$\boxed{x = 16km}$

⑥ A man covered one third of his journey at a speed of 20km/h, one fourth of the remaining at the speed of 25km/h and the remaining journey at a speed of 30km/h ... avg speed of ...

(a) 20km/h (b) 22km/h (c) 24km/h (d) 25km/h (e) 27km/h

LCM of 25 20, 25

$\boxed{300km}$

$\frac{1}{3}$ 200 4 $150\times$

$\frac{100}{2}$ $\frac{50}{25}$ $\frac{150}{30}$

$= 5h + 2h + 5h$ Avg speed $= \frac{Td}{Tt} = \frac{300}{12} = 25$

(7) Ram goes to his school at the rate of 15km/h reach in 15min. late next day he goes with 20 km/h and reached 5 min before then what is distance ~~set~~ between his home to school

(a) 10km (b) 20km (c) 40km (d) 35km (e) NOT

Concept

15km/h → late → 15min

20km/h → early → 5min

Distance = "D" km

Assume @ 10 : late → 10:15] 20 min
early → 9:55

diff late and early then add → 15+5 → 20

if early and early or late and late → subtract 15-5 = 10

$$\left(\frac{D}{15}\right) - \left(\frac{D}{20}\right) = \left(\frac{20}{60}\right) = \text{Ans}$$

s_1 s_2

Formula

$$\boxed{\text{Distance} = \frac{s_1 \times s_2}{s_1 - s_2} \times \text{difference in time}}$$

$$D = \frac{15 \times 20}{5} \times \frac{20}{60}$$

$$\boxed{D = 20km}$$

(8) Due to bad weather condition an aircraft reduce its speed by 200km/hr and reaches 30minute late in flight of 1200km. calculate actual speed of the aircraft in km/h

(a) 900 (b) 600 (c) 650 (d) 850 (e) NOT

$S_1 = S$

$S_2 = (S-200)$

Basic

$$\frac{1200}{S} - \frac{1200}{(S-200)} = \frac{30}{60}$$

$$D = \frac{S_1 S_2}{S_1 - S_2} \times \text{diff time} \quad 1200 = \frac{S(S-200)}{200} \times \frac{30}{60}$$

$$480000 = S \times (S-200)$$

$\boxed{S = 800}$ verify through options

(9) ~~An~~ An ~~a~~ aeroplane is stopped for half an hour and now it has cover a distance of 1500 km in given time. so its speed increased by 250km/h. Find the initial speed in km/h

(a) 1000 (b) 750 (c) 1250 (d) 1500 (e) 500

Initial Speed = S

Now = (S+250)

$$d = \frac{S_1 \times S_2}{S_1 - S_2} \times \text{d int}$$

$$1500 = \frac{S \times (S+250)}{250} \times \frac{30}{60}$$

$$750000 = S \times (S+250)$$

$$\boxed{S = 750km/h}$$

(10) A man cover a certain distance on scooter ~~and~~ if he travels 3km/hr faster. He could have taken 40 min less but if he decrease his speed 2km/h then he becomes 40min late. Find the distance in km

(a) 40 (b) 80 (c) 60 (d) 25 (e) None

1st case | 2nd case

$D = \frac{s \times (s+3)}{3} \times \frac{40}{60}$ | $D = \frac{s \times (s-2)}{2} \times \frac{40}{60}$

$\frac{s(s+3) \times 2}{3 \times 6} = \frac{s(s-2) \times 2}{2 \times 3}$

$2s + 6 = 3s - 6$

$\boxed{s = 12}$

sub in 1st case $D = \frac{12(15)}{3} \times \frac{2}{3}$

$\boxed{D = 40km}$

(11) A train left station A for station B at certain speed. After travelling for 100km, the train meets with an accident and could travel of 4/5th of the original speed and reaches 45min late at station B. Had the accident taken place 50km further on. It would have reached 30min late at station B. What is the distance b/w station A and station B?

original $s = 5x$, after $= 5x \times \frac{4}{5} = 4x$

$\leftarrow s_1 = 5x \rightarrow$ Accident $s_2 = 4x$, 45min late; $\leftarrow 100km \rightarrow$ $\rightarrow 50km \rightarrow$ D_2, 30min late

Let original $s = 5x$

after $= 5x \times \frac{4}{5} = 4x$

$D_1 = \frac{5x \times 4x}{x} \times \frac{45}{60} = 15x$

$D_2 = \frac{5x \times 4x}{x} \times \frac{30}{60} = 10x$

$D_1 - D_2 = 50km$

$5x = 50$

$\boxed{x = 10}$

Total distance $= 100 + 15x$

$= 100 + 15 \times 10$

$= 250 km$

(12) An express train travelled at an average speed of 100km/hr, stopping for 3min after every 75km. How long it take to reach its destination 600km from the starting point? In hours

(a) 6h (b) 6h 24m (c) 6h 21m (d) 6h 30m (e) Not

A —— 600km —— B

$t = \frac{D}{S} = \frac{600}{100} = 6$ hr.

stoppages $= \frac{600}{75} = 8$ but for last stop it will not stop: hence $8 - 1 = 7$

$7 \times 3 = 21$ min

6h + 21min = 6h21m

(13) Relative Speed.

Relative speed: Opposite ($\rightarrow \leftarrow$) $s_1 + s_2$ = Rel speed; Same speed ($\rightarrow \rightarrow$) $s_1 - s_2 = R_s$

(13) The distance between two station A and B is 800km. A train X starts from A and moves towards B at 40km/hr and another train Y starts from B and moves towards A at 60km/hr. How far from A will they cross each other? in km

(a) 330 (b) 320 (c) 300 (d) 360 (e) Not

A $\leftarrow$ 800km $\rightarrow$ B

X (40km/hr) $\rightarrow$ $\leftarrow$ Y (60km/hr)

Concept $= \frac{800}{100} = 8$hr

Relative speed $= t = \frac{800}{100} = 8$hr

40×8

$= 320km$

(d) t = constant A : B

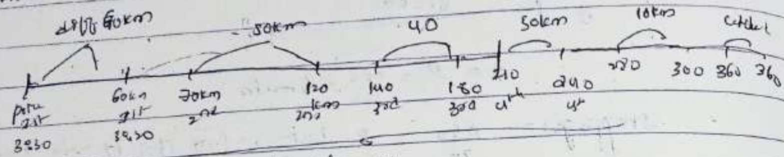
s → 40 : 60

s → d ∝ 2 : 3

d → 2 : 3 → $5p = 800$, $2p = \frac{800 \times 2}{5} = 320km$

(14) A thief steal a car @ 2:30 pm and drive at a speed of 60 km/hr. Police came to know about theft @ 3:30pm and start chasing him with the speed of 70 km/hr. After how much kilometre police will catch the thief?

ⓐ 400 km ⓑ 420 km ⓒ 490 km ⓓ 500 km ⓔ NOT



Concept ⟶ Total 6 hours

Formula $= T = \frac{d}{RS} = \frac{60}{10} = 6$ hours

$d = S \times T = 60 \times 70 = \underline{420\ km}$

(15) Tom spots Jerry 200 m ahead of him and starts chasing him. Jerry realises this after 4 min and starts running away from him. In one leap, Tom covers 3m and Jerry covers 2m. Also, in one min, Tom takes 12 leaps and Jerry takes 16 leaps. In how many min will Tom catch Jerry.

ⓐ 16 min ⓑ 15 min ⓒ 14 min ⓓ 17 min

	Tom	:	Jerry
Tom – 1 leap	3m	:	2m
Jerry – 1 min	12 leaps	:	16 leaps
1 min distance :-	36 m	:	32 m

$RS = 4\ m/s$

Jerry realises after 4 min means (Tom covers dis in 4 min)

1m ⟶ 36m

4m ⟶ $36 \times 4 = 144 = 200 - 144$

$= 56$ metre

$T = \frac{d}{RS} = \frac{56}{4} = 14$ min

(16) A bike during a fog passes a man was walking at a speed of 3 km/hr in the same direction. He could see the bike for 4 min and it was visible to him upto a distance of 100m. Find speed of the bike?

ⓐ 1.5 km/hr ⓑ 4.5 km/hr ⓒ 5 km/hr ⓓ 6 km/hr ⓔ NOT

Bike (S) = ? Man = 3 km/hr same = RS

4 min ⟶ D = 100m

$RS = \frac{d}{t} = \frac{100}{4 \times 60} \times \frac{18}{5} = \frac{3}{2} = 1.5$

$RS = 1.5$ $RS = S_1 - S_2$

$m = 3km$ $1.5 = S_1 - 3$

$\boxed{S_1 = 4.5}$ km/hr

TRAIN PROBLEMS :-

Distance

* (i) point object ⟶ man / pole / Tree

Distance ⟶ own length + {m / p / Tree} negligible compared to train

distance ⟶ own length

* (ii) Platform / Tunnel / Bridge

Distance ⟶ own length + {platform / Tunnel / Bridge}

* (iii) Another train = own length + (length of another length)

On moving objects ex:- man & train

when two bodies are in motion we will not consider individual speed consider the Relative speed

Opposite direction (⇄) ⟶ $S_1 + S_2$

Same direction (⇉) ⟶ $S_1 - S_2$

Conversion ⟶ length

Conversion $\longrightarrow$ $\frac{1 km}{1 hr} = \frac{1000m}{3600m} = \frac{5}{18}$

$km \to m = \cancel{50 \times 8} \; 50 km \times \frac{5}{18} = m/s$

① How many sec will a train of 100m long running @ the rate of 36kmph take to pass the stationary pole

D = own length = 100m

speed = $36 km/h \times \frac{5}{18}$ = 10m/s

$t = \frac{d}{s} = \frac{100}{10} = 10 sec$

② How many sec 110m long train running @ 72kmph take to cross bridge 130m long?

D = own length + Bridge

= 110 + 130m = 240 m

$S = 72 \times \frac{5}{18} = 20 m/s$

$t = \frac{d}{s} = \frac{240}{20} = 12$ seconds

③ Two trains 121m and 99m in length respectively are running opp direction one @ 40kmph and other @ 32kmph. In what time will they completely cross each other from the moment they meet?

T.D = 121 + 99 = 220m

Rs = Opp = $s_1 + s_2$ = 40 + 32 = 72 k/h

$\boxed{t = \frac{d}{s} = \frac{220}{\frac{72 \times 5}{18}} = 11 sec}$

④ Two trains 110m and 90m in length resp. are running in the same direction, one @ 50km/h and other 68k/h. In what time they will cross each other

T.d = 110 + 90m = 200m

Rs = opp.d = 68 - 50 = 18

$\boxed{t = \frac{200 \times 18}{18 \times 5} = 40 \; sec}$

⑤ A train 110m long travels @ 60k/h. In what time will it pass a man who is walking 6k/h

(i) Against it (ii) in the same direction

D = own length = 110

(i) opp.d = $s_1 + s_2$ = 60+6 = 66 k/h $t = \frac{110 \times 18}{66 \times 5} = 6 sec$

(ii) same d = $s_1 - s_2$ = 66 - 6 = 54 k/h = $t = \frac{110 \times 18}{54 \times 5} = \frac{55}{3}$

$= 7\frac{1}{3}$ sec

⑥ Two train are moving in same direction @ 50k/h and 30k/h. The faster train crosses a man in slower train in 18secs. Find the length of the faster train

Train will not cross slower train it crosses a man hence

d = own length of the faster train =

s_1 = 50k/h s_2 = 30 k/h (speed of man) t = 18 sec

$t = \frac{d}{s} = 18 = \frac{d}{50-30}$

$= 18 = \frac{D \times 18}{20 \times 5}$ $\boxed{D = 100m}$

⑦ A 150m long train crosses a platform in 45 sec and cross a man in 9 sec. Find the length of the platform

Train → man D = own length = 150m

$s = \frac{D}{t} = \frac{150}{9}$ = m/s

platform $= x$ $\quad d = 150 + x$ $\quad t = 45 sec$

$t =$

$t = \frac{d}{s}$ $\quad 45 = \frac{150+x}{150} \times 9$

$\boxed{x = 600m}$

or

another method

9 sec → 150m

platform ← 36 sec → $\frac{150 \times 36}{9} = \boxed{600m}$

or

150m → 9 sec

150 + p → 45 sec

p → 36 sec

9 sec → 150

36 sec → $\frac{150 \times 36}{9} = 600m$

(8) A 100m long train crosses a 300m river bridge in 30 sec. In how many sec it will cross a stationary pole

$D = 100 + 300 = 400m$

$t = 30 sec$ $\quad s = \frac{d}{t} = \frac{400}{30} = \frac{40}{3} m/s$

stationary pole ← $t = \frac{d}{s} = \frac{100}{40} \times 3 = 7.5 sec$

(9) A 100m long train crosses another train of length 150m coming from opp. direction in 10 sec. If the speed of the 1st train is 30 k/h find the speed of the second train in k/h

$s_1 = 30 k/h$ $\quad s_2 = x\, k/h$

opp $d = s_1 + s_2 = 30 + x\ k/h$

T.d $= 100 + 150m = 250m$ $\quad t = 10 s$

R. $S = \frac{d}{t} = \frac{250}{10} = 25 m/s \times \frac{18}{5} = 40 k/hr$

$30 + x = 40$

$\boxed{x = 60 k/h}$

(10) Station A and station B are @ 220 km apart. A train leaves station A @ 8:00am @ the 20 k/h and another train leaves station B at 10:00am @ the rate of 25 k/h. At what time will both the trains meet each other.

A → 220m ←

20 k/h $\quad$ 25 k/h

8:00am $\quad$ 10:00am

$d = s \times t = 20 \times 2$

$= 40m$ @ $\quad 220 - 40 = 180m$ @ 10:am $\quad$ ← 180 → $\quad$ 25 k/h @ 10:..

$R.s = 20 + 25 = 45 k/h$

$T.d = 180$ $\quad t = \frac{d}{s} = \frac{180}{45} = 4 hr$

4 hr → 180m + 2 hr → 40m

total 220m → 6 hrs

8+6 → 2:00pm or 10:am + 4h → 2:00 pm

(11) A 120m long train crosses a platform in 63 sec and crosses a railway signal in 27 sec. Find the length of platform

120m → 27 sec

120 + x → 63 sec

$x = 36$

27 sec → 120m

36 sec → ? $\frac{120 \times 36}{27} = 160$

$\boxed{x = 160m}$

AVERAGE :→

$$\text{Average} = \frac{\text{Sum of the entities}}{\text{Total no. of entities}} =$$

23, ~~27~~, 72, 37, 47 → 180

$$Avg = \frac{180}{4} = 45$$

Basic methods → i) Basic method
ii) Sum method
iii) Alligation method
iv) Deviation method

Ex :- The avg weight of 62 students of a school is 54 kg. If the weight of the teacher is included, the avg rises by 3 kg. Find the weight of the teacher.

(a) 186 (b) 179 (c) 243 (d) 234 (e) NOT

Traditional method

Sum = Avg × n

$$\frac{\text{New sum}}{\text{New } n} = \frac{[62 \times 54] + \text{Teacher Avg}}{63} = 57$$

Sum method

n × Avg

62 × 54 → $\boxed{3347}$

↓ +1 ↓ 3

63 × 57 → $\boxed{3591}$

Diff $\boxed{243}$

Alligation method

Existing (Avg) 54 — change (Add/minus) (Avg) x

Finally 57

$x-57$: 3

62 : 1

$$\frac{x-57}{3} = \frac{62}{1} \quad x - 57 = 186$$

$\boxed{x = 243}$

Deviation method :→

n = 10, Avg = 20, Sum = 200

62 → 54

↓ → 57 ⟩ $\boxed{3kg}$

63 → 57

each member increased by 3 then 63×3

54 + 63×3

54 + 189 = 243

↓

Basic of Deviation

n = 10 Avg = 20 Sum = 200

C-1 n = 1 new = 21 $Avg = \frac{200+21}{11} = \frac{221}{11} = 11$

$\boxed{20}+\boxed{1}$

C-2 n = 1 new = 31 $Avg = \frac{200+31}{11} = \frac{231}{11} = 21$

$\boxed{20}+\boxed{11}$

previous avg — each one is increased by $\frac{11}{11} = 1$

C-3 n = 1 new = 9 $Avg = \frac{200+9}{11} = \frac{209}{11} = 19$

200 + 9

$\boxed{20} - \boxed{11}$

① The mean weight of ~~35~~ 34 students of a school is 42 kg. If the weight of the teacher be included, the mean rises by 400 gms. Find the weight of the teacher.

34 → 42

↓ ↓ 400 gm = 0.4 kg

35 42.2

42 + 0.4 × 35

= 42 + 14 = 56 kg

(2) The avg age of 6 workers in a company is 37 years. If a new worker is employed, the avg is decreased by 2 yrs. Find the age of new worker?

(a) 23 kg (b) 56 kg (c) 76 kg (d) 13 kg (e) Not

students

$6 \longrightarrow 37 \longrightarrow \boxed{222}$

$\downarrow \quad \downarrow -2$

$7 \longrightarrow 35 \longrightarrow \boxed{245}$) 23

Deviation method — $37 - 7\times 2$

$37 - 14 = \boxed{23 kg}$

(3) The avg weight of 10 members in a family increases by 2.5 kg when a new member comes in place of one of them weighing 70 kg. Find the weight of new member

(a) 75 kg (b) 75 kg (c) 72.5 kg (d) 95 kg (e) Not

Traditional:→ $\frac{10x + N - 70}{10} = x + 2.5$

$= 10x + N - 70 = 10x + 25$

$\boxed{N = 95}$

Deviation method:→ $70 + 10\times 2.5$

$70 + 25$

$= 95$ kg

Deviation method on level-2

(4) A captain of a cricket team of 11 members is 26 years old and the w.k. is 3 years older. If the ages of these two excluded, the avg age of remaining players is one year less than the avg age of the whole team. Find out the avg age of the team.

(a) 23 (b) 56 (c) 76 (d) 17 (e) Not

$n = 11$ old age $= x$ New A $\longrightarrow x - 1$

$C = 26$ $w = 29 \rightarrow 55$

$\frac{11x - 55}{9} = x - 1$

$= 11x - 55 = 9x - 9$

$= 2x = 55 - 9 = 46 \Rightarrow \boxed{x = 23}$

Deviation:→ Assumption → Avg

$55 - 9 = 46 = 46/2 = 23$

(5) The avg age of 30 students of a class is 15 years. A student of 20 years left the class and two more students, whose age difference was 5 years, joined the class in place of him. If the avg of the class was still 15 yrs then find the age of the younger new student!

(a) 43 (b) 56 (c) 15 (d) 17 (e) Not.

Basic

n	Avg	Sum
30	× 15	= $\boxed{450}$
−1 +2		−20 → $\boxed{430}$ → 35 (2 members)
$\boxed{31}$	× 15	= $\boxed{465}$

$\boxed{20}$ $\boxed{15}$

$x + x - 5 = 35$

$2x = 40$

$\boxed{x = 20}$

$\boxed{y = 15}$

Deviation :– Avg = 15

−20 left → (−5) → extra

$2 \longrightarrow 2\times 15 = \boxed{30} + \boxed{5} = \boxed{35}$

(6) A batsman played 40 innings with avg score of 50 runs. His highest score exceeds his lowest score by 172 runs. If these two innings are excluded, then avg score of

his remaining innings in 40 runs. what is the highest score of this match?

ⓐ 132 ⓑ 156 ⓒ 174 ⓓ 132 ⓔ NOT

Basic

n	×	Avg	=	
40	×	50	=	2000
38	×	47	=	1824
(43−5)		(43+5)		

$2000 - 1824 \rightarrow 176$

H $\boxed{174}$ L $\boxed{172}$

Advance (Deviation)

Assumption: Avg = 50

~~H+L = 50×2~~ H+L = 50×2 + 38×2

= 100+76 → $\boxed{176}$

Deviation on Level-3 :→

⑦ Average weight of m boys 43 kg if the weight of their teacher who weighs 63 kg also included then avg becomes 45 kg. Find the value of m

ⓐ 6 ⓑ 8 ⓒ 9 ⓓ 10 ⓔ Not

n = m Avg = 43, sum = 43·m

n	×	Avg	=	
m	×	43	=	43m
m+1	×	45	=	45(m+1)

$\frac{43m+63}{m+1} = 45$

$45m + 45 = 43m + 63$

$2m = 18$

$\boxed{m = 9}$

$\boxed{43} \rightarrow \boxed{63}$

$\frac{20}{2} = 10 - 1 = 9$ students

⑧ The average age of some students is 25. A teacher joins them of age 56.5 so that the new age avg of all students becomes 26.5 find out how many students are there.

ⓐ 21 ⓑ 20 ⓒ 16 ⓓ 18 ⓔ Not

$\frac{25m + 56.5}{m+1} = 26.5$

$25m + 56.5 = 26.5m + 26.5$

$1.5m = 30 \quad m = \frac{30}{1.5} = \boxed{m = 20}$

Assumption :— Avg = 25

25 → 56.5

$\frac{31.5}{1.5} = 21 \rightarrow$ included teacher = 20 students

⑧ A batsman in his 12th inning makes a score of 63 runs by increases his avg score by 2. what is his avg after 12th inning

ⓐ 39 ⓑ 41 ⓒ 36 ⓓ 44 ⓔ Not

$\frac{11x + 63}{12} = x + 2$

$= 11x + 63 = 12x + 24$

$\boxed{x = 39}$ $\quad \frac{39 + 63}{12} = \frac{102}{12}$

After 12th = 39+2 = 41

Advance (Deviation)

Extra

12th → $\boxed{63}$ ~~24~~ 12×2

63 − 24 = 39 + 2 = 41

⑩ Multiple Average problems

⑨ The avg of 12 no is 20. The avg of 1st 5 no is 18. and that of last 6 no. is 22. find the 6th no.

ⓐ 22 ⓑ 16 ⓒ 26 ⓓ 18 ⓔ Not

Ans $12 \times 20 = 240$

1^{st} 5 = $5 \times 18 = 90$ → 222

2^{nd} 6n = $6 \times 22 = 132$

$240 - 222$

$= \underline{18}$

⑨ Deviation :-

$\boxed{12} \longrightarrow \boxed{20}$

$\boxed{5} \to \boxed{18}$ $\boxed{6} \to \boxed{22}$

$20 - 18 = -2 \times 5$ $22 - 20 = +2 \times 6$

$= \boxed{-10} \longleftrightarrow^{+2}$ $= \boxed{12}$ $20 - 2 = \boxed{18}$

⑩ In a series of 11 matches of cricket, avg score of Rohit Sharma is 85. In first 4 matches his avg score was 72 and in next 6 matches → 90. Avg. How many runs scored by Rohit Sharma in last match.

ⓐ 115 ⓑ 110 ⓒ 107 ⓓ 97 ⓔ 112

Basic

$11 \times 85 = 935$

$11 \left[\begin{array}{l} 4 \times 72 = 288 \\ 6 \times 90 = 540 \end{array} \right] \to 828$

$935 - 828$
$= \underline{107}$

Adva (Deviation)

$11 \longrightarrow \boxed{85}$

$4 - \boxed{72}$ $6 - \boxed{90}$

$-13 \times 4 =$ 5×6

$\boxed{-52} \xrightarrow{-22} \boxed{+30}$

Last match = $85 + 22$
$= \underline{107}$

⑪ Average of 25 no's is 67. Average of first 10 no's is 64 and that of last 13 no's is 69. If the 11th no' is 25% of 12th no, then find the 12th no.

ⓐ 27.6 ⓑ 135 ⓒ 108.8 ⓓ 110.4 ⓔ NOT

Basic

$25 \times 67 = 1675$

$10 \times 64 = 640$

$13 \times 69 = 897$ 1537

11, 12

138

① ④

$\frac{138}{5}$ 27.6 = 27.6×4
= 110.4

Advance (Deviation) → Base

67

$10 \to \boxed{64}$ $13 \to \boxed{69}$ $\boxed{2} \to$?

-3×10 2×13 $2 \times 67 = 134$

$= \boxed{-30}$ $\boxed{26}$ $+ 4$

-4 $138 - 2$ no's

⑫ The average of 8 observations was 25.5. It was noticed later than that two of those observations were wrongly taken. One observation was 14 more than the original value and other observation was wrongly taken as 31 instead of 13. What will be the correct average of those 8 observation?

ⓐ 21 ⓑ 21.5 ⓒ 22.5 ⓓ 29.5 ⓔ NOT

Basic

$8 \times 25.5 = 204$

Original $\longrightarrow \boxed{-14}$

$31 \longrightarrow 13 \longrightarrow \boxed{-18}$

Total $\longrightarrow \boxed{-32}$

Correct → $204 - 32 = 172 \longrightarrow \frac{172}{8} = \boxed{21.5}$

Deviation (Advance)

25.5 $\frac{-32}{8} = -4$

$\underline{21.5}$

(13) Average temp of the place in 25 days is recorded at 15°c. Later it was found that average temp of four days which was 14°c by mistake taken as 16.5°c. Find the Original average temp of that place within that duration.

(a) 14.8°c (b) 14°c (c) 14.4°c (d) 14.6°c (e) 14.2°c

Basic

25 × 15 → 375

Correct → 14° → Incorrect → 16.5

2.5 × 4 = $\boxed{10°c}$

Correct → 375 − 10 → $\frac{365}{25}$ = 14.6°c

Advance (Deviation)

$\boxed{15}$ $\frac{2.5 \times 4}{25} = \frac{10^{2}}{25_{5}} = \boxed{-0.4}$

$\boxed{14.6}$

(14) The average marks of 14 students was calculated as 71. but later it is found that there was an error in noting the marks of 2 students as 42 instead of 56 and 74 instead of 32. What will be the correct average of the student?

(a) 53 (b) 66 (c) 69 (d) 59 (e) NOT.

Basic

14 × 71 → 994

Correct → 56 → 72

Incorrect → 42 74

(+14) (−42) → $\boxed{-28}$

994 − 28 → $\frac{966}{14}$ = $\boxed{69}$

Advance (Deviation)

71 $\frac{-28}{14}$ −2

→ $\boxed{69}$

Alligation methods

(15) The avg marks of 21 students in a class is 42. If 15 students with an average mark of 78 join them then find the avg marks of all the students

(a) 53 (b) 56 (c) 57 (d) 58 (e) Not.

Existing — 36 — Add (new)

$\boxed{42}$ $\boxed{78}$

New avg $\boxed{x}$

$\boxed{78-x}$: $\boxed{x-42}$

21 : 15

7 : 5

21 : 15

7 : 3

$\boxed{21}$ $\boxed{15}$

$\frac{78-x}{x-42} = \frac{7}{5}$

$\boxed{x = 57}$

(16) The mean weight of 180 students in school is 50 kg. The mean weight of boys is 60 kg and that of girls is 45 kg. Find the no. of boys and girls in the school.

(a) 100 & 80 (b) 60 & 120 (c) 70 & 120 (d) 150 & 30 (e) NOT

Boys girls

$\boxed{60}$ $\boxed{45}$

$\boxed{50}$

5 10

1 : 2

3 → 180

$\boxed{1 = 60}$ 60 : 120

Basic

B = x g = y

Avg = 60 Avg = 45

sum = 60x sum = 45y

$\frac{60x + 45y}{x+y} = 50$

⇒ 10x = 5y

$\frac{x}{y} = \frac{5}{10} = \frac{1}{2}$

60 : 120

(17) Average weight of class is 60 kg and avg weight of boys in the class is 70 kg. Ratio of boys to girls in the class is 5 : 4. If there are 72 students in the class, then find avg weight of girls in the class.

(a) 54 kg (b) 42 kg (c) 35 kg (d) 45 kg (e) 38 kg

Boys : Girls

[80] : [x]

[60]

[60−x] : [20] } 5

5 : 4

$\frac{60-x}{20} = \frac{5}{4}$

$240 - 4x = 100$

$140 = 4x$

$\boxed{x = 35}$

(17) The average salary of all the workers in a workshop is ₹ 8000. The average salary of 7 technicians is ₹ 12,000 and the average salary rest is 6000. the total no. of workers in the workshop is

(a) 14 (b) 22 (c) 21 (d) 30 (e) NOT

Basic :- Technician = 7 → 12000 = 84000

Rest = x → 6000 = 6000x

Whole = $\frac{84000 + 6000x}{x+7} = 8000$

= 84000 + 6000x = 8000x + 56000

28000 = 2000x

$\boxed{x = 14}$ Total workers x+7 = 14+7 = 21

Tech (Avg) 12,000 Rest (Avg) 6000

Whole Avg 8000

2000 4000

Technicians 1 : 2 → 14

7+14 = 21 workers

Formula based problems :→

Avg Fr — Sum — Avg

First 'n' natural no's → $\frac{n(n+1)}{2}$ $\xrightarrow{\div n}$ $\frac{(n+1)}{2}$

" " odd no's → n^2 $\xrightarrow{\div n}$ n

" " Even no's → $n(n+1)$ $\xrightarrow{\div n}$ $n+1$

" " perfect squares → $\frac{n(n+1)(2n+1)}{6}$ $\xrightarrow{\div n}$ $\frac{(n+1)(2n+1)}{6}$

" " perfect cubes → $\left[\frac{n(n+1)}{2}\right]^2$ $\xrightarrow{\div n}$ $\frac{n(n+1)^2}{4}$

(19) Find the avg of 1st 18 natural no's!

$\frac{n+1}{2} = \frac{18+1}{2} = \frac{19}{2} = 9.5$

(20) Find the avg of first 20 multiple of "3"

Multiples of 3 → 3+6+9+12+15+ ... 20

= 3(1+2+3+4+5+6 ... 20)

= $3\left(\frac{n+1}{2}\right) = 3\left(\frac{20+1}{2}\right) = 3\left(\frac{21}{2}\right) = 3 \times 10.5$

= 31.5

(21) If the avg of first "n" consecutive odd no's is 60 then find the sum of number

avg of n consecutive odd no's is n hence n=60

Sum = $n^2 = (60)^2 = 3600$

(22) Calculate the avg of the square of natural no's 1 to 10

Avg = $\frac{(n+1)(2n+1)}{6} = \frac{11 \times (2 \times 10+1)}{6} = \frac{11 \times 21}{6} = \frac{77}{2}$

= 38.5

(23) Calculate the avg of the cubes of 1st 10 natural no's

Avg = $\frac{n(n+1)^2}{4} = \frac{10(11)^2}{4} = \frac{5 \times 121}{2}$

= $\frac{605}{2}$

= 302.5

(24) Avg of 27 consecutive even no is 82. Then find the largest and smallest numbers

First → x
next → $x+2$

Formula
Highest no → Average + (n−1)
lowest no → Average − (n−1)

Highest → 82 + (26) = $\boxed{108}$
Lowest → 82 − (26) = $\boxed{56}$

(25) Avg of 14 consecutive odd no is 38. Then find the largest and smallest no.

same as above

Largest → 38 + (13) = $\boxed{51}$
Lowest → 38 − 13 = $\boxed{25}$

Average of Arithmetic progression →

(26) Avg of 13 consecutive multiples of 11 is 14641. Find the second largest no of the series

(a) 14652 (b) 16492 (c) 14696 (d) 14607 (e) NOT.

Basic → $\frac{x, x+11, x+22, x+33, x+44 + \dots}{13} = 14641$

A.P → sequence of number → diff between term will become

→ Avg consider term

20 25 $\boxed{30}$ 35 40
gaps: 5 5 5 5

5×(11) second largest

1 2 3 4 5 6 $\boxed{7}$ 8 9 10 11 $\boxed{12}$ $\boxed{13}$ → largest
gaps: 11 11 11 11 11 11 11 11 11 11 11 11

Avg 7th term = 14641
↓
14641 + 5×11 = 14641 + 55
= $\boxed{14696}$

(27) Avg of 9 con multiples of 17 is 49189.5 Find the second smallest no of the series

(a) 49197 (b) 49497 (c) 49147 (d) 49749 (e) NOT

+8.5 $\frac{17}{2} = 8.5$ −8.5

middle term

1 2 3 4 5 6 7 8
gaps: 17 17 17 17 17 17 17

Avg
49189.5 − 8.5 ~~17 17~~

4th term = 49189.5 − 8.5 = 49181
2nd term = 49181 − 17 − 17
= $\boxed{49147}$

(28) S1 is the series of 5 consecutive multiple of four, whose sum is 100 and S2 is the series of four consecutive even no, If the second no of S2 series is 6 less than highest no of S1 series, then find avg of S2 series, If S2 is consecutive even number series?

(a) 22 (b) 23 (c) 25 (d) 21 (e) NOT

S1 → ☐ ☐ $\boxed{20}$ ☐ ☐ (24, 28)
gaps: 4 4 4 4
Avg = 20
$\frac{100}{5} = 20$
Avg

S2 → ☐ ☐ ☐ ☐ (20, 22, 24, 26)
−6
gaps: 2 2 2

$\boxed{Avg = 23}$ final

(29) The avg run/wicket of a bowler is 24.85. In the next match he gives 52 run and take 5 wicket and average improve by 0.85. Find the total no of wicket he has taken

(a) 80 (b) 85 (c) 75 (d) 90 (e) NOT

Bowling Avg = $\frac{\text{Total run conceded } (24.85x)}{\text{Total wicket } (x)}$

$\frac{24.85x + 52}{x+5} = 24$ (24.85 − 0.85)

$24.85x + 52 = 24x + 120$

$0.85x = 68$ $x = \frac{6800}{85} = 80$

Total wicket = x+5 = 85

(30) A man, a woman and a boy together complete a work in 3 days. If a man alone can complete it in 6 days and a boy alone completes it in 18 days, then in how many days a woman will be able to finish the work

(a) 9 (b) 21 (c) 24 (d) 27

1M + 1W + 1B → 3 days

1M → 6 d

1B → 18

18: (6), (3), (1)

M + W + B = 6

3 + W + 1 = 6

$\boxed{W = 2}$

$\frac{18}{2} = \boxed{9}$

(31) A man and a boy together can comp

TIME AND WORK

$$\boxed{\text{Time} = \frac{\text{Work}}{\text{Efficiency}}}$$

Ex:→ A can do a piece work in 20 day and B can alone can do it in 30 day. In how many days A & B can complete the work

total A → 20 days total → 60 work
B → 30 days

Basic →
Time required to complete work = A = 20

A's 1 day work = $\frac{1}{20}$

Iilly B's " " = $\frac{1}{30}$

Together = $\frac{1}{20}+\frac{1}{30} = \frac{3+2}{60} = \frac{5}{60} = \frac{1}{12}$

A & B together do it in = 12 days

Efficiency method :→

Time	T.W	Efficiency
A → 20 days	60 unit	$60/20 = 3$ units/day
B → 30 days		$60/30 = 2$ units/day

$T = \frac{W}{E} = \frac{60}{5} = 12$

① A and B together can do it a piece of work in $7\frac{1}{2}$ day and B alone can do it in 25 days. In how many days A alone finish the work

ⓐ $10\frac{5}{7}$ ⓑ $10\frac{4}{7}$ ⓒ 10 ⓓ Not

A+B = $7\frac{1}{2}$ = $\frac{15}{2}$ days → 75 → $\frac{75}{15/2} = 10$ u/d

B → 25 days → $75/25 = 3$ u/d

numerator's LCM = 75

A+B = 10 B = 3 10 − 3 = 7 → A

A's Time = $\frac{75}{7} = 10\frac{5}{7}$

② A, B and C can do a piece work in 8 days. If B alone do the same work in 18 days and C alone can do it same work in 24 days. In how many days A alone can finish the same work.

ⓐ 30 ⓑ 32 ⓒ 36 ⓓ 40

	T	T.W	E
A+B+C →	8	144	18
B →	18		8
C →	24		6

A's E = 18 − 14 = 4

$\frac{144}{4}$ = 36 = 36 days

③ A can do ½ of a piece of work in 5 day, B can do 3/5th of the same work in 9 days. and C can do 2/3 of that work in 8 days. In how many days can three of them together do the work

ⓐ 3 ⓑ 5 ⓒ $4\frac{1}{2}$ ⓓ 4

	T	T.W	E
A = $\frac{1}{2}$ → 5 days →	10 days	60	6
B = $\frac{3}{5}$ → 9d →	15 d		4
C = $\frac{2}{3}$ → 8d →	12 d		5

6+4+5 = 15 $\frac{60}{15}$ = 4 days.

④ A and B do a work in 8 days, B and C do it in 24 days and C & A it in $8\frac{4}{7}$ days. how many days C alone can do the whole work

ⓐ 60 ⓑ 40 ⓒ 30 ⓓ 10

	T	T.W	E
A+B →	8d	120	15
B+C →	24		5
A+C →	$\frac{60}{7}$		$\frac{120 \times 7}{60} = 14$

$8\frac{4}{7} = \frac{60}{7}$

34 units

$2(A+B+C) = 34$

$A+B+C = 17$ where $A+B = 15$ hence $C = 2$

$C's\ Time = \frac{120}{2} = 60$ days

Man Leaving & Joining problems

(5) A can complete a piece of work in 20 days and B can complete the same work in 25 days. If they start the work together but after 5 days B left the work. In how many days the total work would be finished.

(a) 12 (b) 15 (c) 16 (d) NOT

A → 20 d, B → 25 d → T.W 100 units → E: 5, 4

$A+B = 5+4 = 9$ 5 days w of $A+B = 9\times5 = 45$

$100-45 = 55/5 = 11$ days Total w.k = $11+5 = 16$ days

(6) A and B can do a piece of work in 45d and 40d respectively. They began to do work together but A leaves after some days and B completed remaining work in 23 days. Then after how many days A left the work?

(a) 9 (b) 12 (c) 13 (d) 15

A → 45d, B → 40d → 360 units → 8, 9

B's w in 23 days = $23w = T\times E = 23\times9 = 207$ unit

$360-207 = 153$ 153 w completed by A+B $= \frac{153}{9\times8}$... $\frac{153}{17} = \boxed{9}$

A left after 9 days.

(7) A and B can do a piece of work in 30 days, while B and C can do the same work in 24 d and C and A in 20 days. They all work together for 10 days when B and C leave how many days more will A take to finish the work!

(a) 17 (b) 20 (c) 21 (d) 24

A+B → 30, B+C → 24, A+C → 20 → T.W 120 units → 4, 5, 6

$A+B + B+C + A+C = [2(A+B+C) = 15]$

$A+B+C = 7.5$

A+B+C w together 10 d = $7.5\times10 = 75$ $120-75 = \frac{45}{2.5} = \frac{450}{25} = 18$

A+B+C = 7.5, 4+C = 7.5; A+5 = 7.5, $\boxed{A=2.5}$

$\boxed{A's\ time = 18\ days}$

(8) A can do a work in 16 days, B can do it in 36d and C can do it in 48d. A, B and C start working together after working 4d A left then after 5 days B also left then in how many days C complete the remaining work?

(a) 10 (b) 12 (c) 15 (d) NOT

A → 16d, B → 36d, C → 48d → T.W 144 units → 9, 4, 3

4d's w of A+B+C = $16\times4 = 64$ 5d's w of B+C = $7\times5 = 35$

$64+35 = 99$ $144-99 = 45$ $\frac{45}{3} \to$ C's Time $= \boxed{15}$ d

☆ (9) A, B and C can do a work and 10d, 12d, and 15d respectively. They start work together but A left after some working 3d and C left 4d before the completion of the work. In how many much time the whole work completed.

(a) $6\frac{4}{9}$ (b) $4\frac{7}{9}$ (c) $7\frac{5}{9}$ (d) $6\frac{8}{9}$

10d – A, 12d – B, 15d – C → 60 units

$6\times3 = 18$

$5\times x = 5x$

$4\times(x-4) = 4x-16$

$4(x-4) + 6\times3 + 5x = 60$

9. $18 + 15x + 4x - 16 = 60$

$9x = 58 \Rightarrow x = \frac{58}{9} = \boxed{6\frac{4}{9}}$ d

(10) P takes 10 d to complete 40% of a task while Q takes 18 d to complete 75% of the same work. P+Q work together but P leaves the work after few days, Q continued to work alone for 3 days. If 63% work is still left, how many d did P leave the work?

(a) 1 (b) 2 (c) 3 (d) 4

$P \rightarrow 40\% = \frac{2}{5} \rightarrow 10 \longrightarrow \boxed{25}$ $Q \rightarrow 75\% = \frac{3}{4} \rightarrow 18 \longrightarrow \boxed{24}$

Total 100%.

$40\% \rightarrow 10$

$100\% = \frac{10 \times 100}{40} = 25$

$P \rightarrow 25d$, $Q \rightarrow 24d$ → 600 units ← 24, 25

$P+Q = 24+25 = 49$ Let P+Q work together through x days

then $49x + 25 \times 3 = 37\%$ ~~100~~ T.W

$49x + 75 = \frac{37}{100} \times 600$

~~$49x + 75$~~ $49x = 222 - 75 = 147$

$\Rightarrow 49x = 147$

$\boxed{x = 3}$

By working P+Q x=3 days P left the work

Alternate days Concepts

(11) A, B and C do work in 20d, 25d and 30d respectively. In how many day will the work be finished. If they do it alternately. A begin the work!

(a) 24 (b) $24\frac{2}{3}$ (c) $24\frac{4}{15}$ (d) Not

A → 20 d, B → 25 d, C → 30 d → 300 units ← 15, 12, 10

37 units: A, B, C = 15, 12, 10 → 37

1st 2nd 3rd 4th 5th 6th: A B C → 37

→ ~~6 days~~ → 74 ''

$\frac{9}{}$ → 226

Take multiples of 37 nearest to 300

$\times 8$: 37 units → 3d, 296 unit → 24 d

Still 4 units → $\frac{4}{15}$ Ans

$24 + \frac{4}{15}$

$\left(24\frac{4}{15}\right)$

(12) A, B and C can do a piece work in 11d, 20d and 55d respectively. If A work continuously and B and C work alternatively with A. In how many days total work will be finished

(a) 7 (b) 8 (c) 9 (d) 10

A → 11d, B → 20d, C → 55d → 220 units ← 20, 11, 4

1	2	3	4	← days
A+B	A+C	A+B	A+C	
20+11	20+11	20+11	20+11	
31	24	31	24	

$\underbrace{31 + 24}_{55}$ $\underbrace{31 + 24}_{55}$

$\times 4$: 55 units → 2 days, 220 units → 8 days

(13) A can do a piece of work in 30 days, B can do in 45 days and C can do same work alone in 60 days. If on the 1st day A worked alone and on the second day A and B work together and on the 3rd day A and C worked together. If they repeat the cycle as follows, then in how many days total work can be completed?

(a) 21d (b) $25\left(\frac{5}{6}\right)$ d (c) $21\left(\frac{5}{6}\right)$ d (d) 24 d (e) Not

A → 30d, B → 45d, C → 60d → [180 units] → 6, 4, 3

1st	2nd	3rd
A	A+B	A+C
6	6+4=10	6+3=9

25

7 (25 units → 3 days) ×7
175 units → 21 days

5 units → 5/6

21 + 5/6 = $\left(21\frac{5}{6}\right)$ days

Efficiency Concepts

(14) If the working efficiency of A is 50% more than B. If B alone can do a piece of work in 36 days. In how many days A alone finish the same work

(a) 15 (b) 20 (c) 24 (d) 30

$E \propto \frac{1}{T}$

	A	:	B
E →	150	:	100
E →	3	:	2
T →	2	:	3

$\frac{1}{2}$ B = 2, A = 3 ⇒ 2+1 = 3

3 → 3 unit × 12 → [36d]

3 unit (B's time) = 36 d

2 unit (A's time) = $\frac{36 \times 2}{3}$ = [24 days]

(15) A is thrice as good as B is work. A is able to finish a work in 24d less than B. In how many days they are together finish the whole work?

(a) 9 (b) 12 (c) 15 (d) 18

	A	:	B
E →	3	:	1
T →	1	:	3

2p → 24d ⇒ 1p = 12d

~~4p → 24×4/2~~ =

T.W = 12×3 = [36] $\frac{36}{4}$ = 9 days

(16) Sam and Ravi together can paint a house in 5 days. If Ravi is 20% less efficient than Sam and Saran who is 20% more efficient than Sam, then find the no of days taken by Ravi and Saran together to paint the house

(a) 4d (b) 4.5d (c) 6d (d) 7.5d (e) None

	R	Sam	Saran
	A	B	C
E →	5	4	6

5d

$-\frac{1}{5}$ = 4 : 5 (Ra : Sam)

$= \frac{1}{5}$ = 5 : 6 (Sam : Saran)

T.W = 9×5 = 45 units

$\frac{45}{10}$ = [4.5d]

(17) A and B can complete a work in 18d where B and C together can complete the same work in 15d. If C is 50% more efficient than A then find the time taken by A alone to complete the whole work?

(a) 30 (b) 45 (c) 50 (d) Not

A+B−B+C = (A+C)

variation due to A & C

A+B → 18d | T 6
B+C → 15d | 5

E 5, 6 — 50% → 1 unit; A's 100% → 2 unit; C's 150% → 3 unit

A → 2, B → 3, C = 3

A+B = 5
2+B = 5
[B=3]

T.W = 2+3 = 5×18 = 90

90/2 = 45d

(18) The efficiency A is 20% more than the efficiency of C and the efficiency of D is 25% less than the efficiency of C. If A and B together can complete the work in 14d and C and D complete the work together in 36d, in how many days B alone can complete the work?

(a) 42d (b) 45d (c) 49d (d) 40d (e) 39d

$A:C = 1 \cdot \frac{A}{C} = \frac{3}{1}$ $\frac{C}{D} = \frac{3}{4}$

A : B : C : D

4(3 : : 1 : 4

12 : : 4 : 3

14d 36d

$4+3 = 7eff \to 36$

$T.W = 36 \times 7 = 252$

$T.W = \frac{252}{14d} = \frac{36 \times 7}{14} = 18$

$A+B\ eff = \frac{252}{14} = \boxed{18}$

$\boxed{12} + \boxed{6} = \boxed{18}$

B'eff =

$\text{Time of B's} = \frac{252}{6} = \boxed{42}$

(19) Time taken by A to complete the job is 80% more than time taken by A and B together to complete the same job. Find the time taken by A in days to complete the job alone if C is 100% more efficient than B and time taken by B and C together to complete the job is 20/3 days.

(a) 20d (b) 16d (c) 18d (d) 12d (e) NOT

$\frac{A}{A+B} = \frac{9}{5}$ A : A+B

E = 9 : 5

t = 5 + 9

$A+B = 9$

$5+B = 9$

$\boxed{B=4}$

C is 100% than B hence $\boxed{C=8}$

$B+C = 8+4 = 12$ $T.W = T \times E = 12 \times \frac{20}{3} = 80$

A's time $\frac{80}{5} = ~~12\ days~~$ 16days

Man - women - ~~Days~~ hour - Concepts

(20) 15 men working 8h/d make a dam in 10 days in how many days will 12 men working 10 h/day will make it

(a) 8 (b) 10 (c) 9 (d) 12

M.W = Man × days × Time

$M_1 \times D_1 \times T_1 = M_2 \times D_2 \times T_2$

$15 \times 10 \times 8 = 12 \times D_2 \times 10$

$\boxed{D_2 = 10}$

(21) 24 labourers working 9h/d make a road in 15 days. 48 labourers working 6h/d in how many days will make the triple long road?

(a) 20d (b) 32d (c) 30d (d) 15d

$M_1 \times d_1 \times T_1 = M_2 \times d_2 \times T_2$

$24 \times 15 \times 8 \times 3 = 48 \times 6 \times D_2$

$\boxed{D_2 = 30\ days}$

(22) 18 women complete a work in 16 days. they started the work together and after x days, 6 women left the work. Remaining work is completed by 12W ~~and~~ in 12 days. Find the value of x

(a) 6 (b) 7 (c) 9 (d) 8 (e) NOT

$18 \times 16 = 18x + 12 \times 12$
(T.W)

$288 = 18x + 144$

$18x = 144 \Rightarrow \boxed{x=8}$

(23) A contractor contracted to finish a work in 60 days. He employed 32 men to work. But 1/3rd of work was done in 30 days. How many additional men would be employ that remaining work be finish in duration? (a) 38 (b) 30 (c) 32 (d) 24

$M \times D = 32 \times 30 = 960$

$T.W \times \frac{1}{3} = 960$

$\Rightarrow T.W = 960 \times 3$

$\text{Remaining} = 960 \times 3 \times \frac{2}{3} = (32+x) \times 32$

$\Rightarrow 60 = 32 + x$

$\boxed{x = 28}$

$960 \times 3 \times \frac{2}{3} = (32+x) \times 30$

(24) 7 men & 2 boys & 3 girls can do piece of work in 88 days. In how many days one man one day, and one girl. In how many days can do the same work

(a) 45 (b) 48 (c) 42 (d) 60

$1\,man = 2\,boys = 3\,girls$ → Reciprocals

$M : B : G$

$E \to \left(\frac{1}{1} : \frac{1}{2} : \frac{1}{3}\right) \times 6$

$E \to 6 : 3 : 2$

$7m + 2b + 3g \to 6+3+2 = 11$ $= \frac{T.W}{E} = \frac{528}{11} = \boxed{48}$

$T.W = 1\,man = 88\,day$

$88 \times 6 = 528$

(25) 6 men and 8 boys do a work in 10 days while 26 men and 48 boys do it in 2 days. In how many days will 15 men and 20 boys do that work?

(a) 4 (b) 5 (c) 6 (d) 7

$w = \boxed{M \times d}$

$(6M + 8B) \to 10\,days$

$(26M + 48B) \to 2\,days$

$(6M + 8B) \times 10 = (26M + 48B) \times 2$

$30M + 40B = 26M + 48B$

$4M = 8B$

$\frac{M}{B} = \frac{8}{4} = \frac{2}{1}$

$T.W = (6M + 8B) \to 10 \quad (12+8) \times 10 = 200$

Efficiency ⇒ $15M + 20B \to 30 + 20 = 50$

$15M + 20B$ work $= \frac{200}{50} = 4$

(26) Unique ~~word~~ word problems

(26) A alone would take 16 days more to complete a work than A&B together. B take 4 days more to complete a work alone than A&B work together. In how many days A alone can do it.

(a) 20 (b) 24 (c) 26 (d) 42

Derivation = If $A+B = t$, $A = t+a$, $B = t+b$

$A = t+a$, $B = t+b$ → $(t+a)(t+b)$

$Time = \frac{(t+a)(t+b)}{t+b+t+a} = t$

$t^2 + bt + t^2 + at = t^2 + tb + at + ab$

$t^2 = ab \Rightarrow \boxed{t = \sqrt{ab}}$

Let $A+B = t$

$A = t+16$

$B = t+4$

$t = \sqrt{ab} = \sqrt{16 \times 4} = \sqrt{64} = 8$

$A = 16+8 = \boxed{24}$ $B = 8+4 = \boxed{12}$

(27) A can complete a work in 5 more days than B, and does the same work in 9 more days than C. If A and B can complete the whole work in same time in which C alone does the whole work. In how many days A alone complete the same work.

(a) 15 (b) 17 (c) 19 (d) 20

$A+B \to C$

① $A = B+5$

② $A = C+9$

from ① $B = A-5 = C+9-5 = \boxed{C+4}$

$t = \sqrt{ab} = \sqrt{9 \times 4} = 3 \times 2 = 6$

hence Ans $= C+4 = 6+4 = \boxed{10}$

(28) A and B together can do a piece of work in 12 days which B and C together do in 16 days. If A work for 5 days, B work for 7 days then C complete the remaining work in 13 days. In how much time B alone do whole work?

(a) 48 (b) 24 (c) 16 (d) 12.

T.W E

$A + B \to 12\,d$ > 48 units < 4, 3

$B + C \to 16\,d$

A B C

5 7 13

5 2 2 11

$A+B = 4\times5 = 20$] 26 $48-26 = 22$

$B+C = 3\times2 = 6$ $C = \frac{22}{11} = 2$

After substitution ⇒ $B=1$ $C=2$ $A=3$ $\frac{47}{1} = \boxed{47\ days}$

29th in Avg

(30) A man and boy together can complete a work in 24d. If man works alone in last 6 days, then work is finished in 26d, How much time will the boy alone be able to finish his work.

(a) 72 (b) 20 (c) 24 (d) 36

$24M + 24B = 26M + 20B$

$\Rightarrow 2M = 4B \Rightarrow \frac{M}{B} = \frac{4}{2} = \frac{2}{1}$

$T.W = \frac{72}{1} = \boxed{72}$

efficiency Boy = 1

(31) To do a piece of work B takes 3 times as long as A and C together and C twice as long as A and B together. If all the three together can complete the work in 10 d. How long would C complete whole work

(a) 30 (b) 25 (c) 35 (d) 40

Time → $\frac{B}{A+C} = \frac{3}{1}$ $\frac{C}{A+B} = \frac{2}{1}$

$E \to \frac{B}{A+C} = \frac{1}{3}$]+ $\frac{C}{A+B} = \frac{1}{2}$]+   LCM of 3 and 2 is 12 hence

$A+B+C = 4$ ~~$A+B+C = 3$~~

$\frac{12}{3}$ 4

$\frac{B}{A+C} = \frac{4}{8}$ $\frac{C}{A+B} =$

Leading to equal ratios LCM of 3, 4 = 12 hence

$\frac{B}{A+C} = \frac{3}{9}$ $\frac{C}{A+B} = \frac{4}{8}$

$A=5,\ B=3,\ C=4$ $A+B+C = 12\times10 = 120$

$C = \frac{120}{4} = 30$ days

previous year

(32) A, B, C together work for $5(\frac{5}{47})$ days to complete a job. A is 33 1/3 % more efficient than C and C and B together can complete a job in $8(\frac{8}{9})$ days. Find the time taken by C to complete the job.

(a) 14 days (b) 20 days (c) 16 days (d) 12 days (e) NoT

$A+B+C = 5(\frac{5}{47}) = \frac{240}{47} \to \frac{W}{E}$

$B+C = 8(\frac{8}{9}) = \frac{80\times3}{9\times3} \to \frac{240}{27} \to \frac{W}{E}$

$A = 47-27 = 20$ $C = 15$ $B = 12$

33 1/3 % $= \frac{1}{3}C$ $\frac{240}{15} = \boxed{16}$

(33) B is 20% more efficient than A. B started the work to do it for x days, then B is replaced by A. And A completed the remaining work in (x+8)d, Ratio of workdone by A & B is 3:2. In how many day A & B working together to complete the whole work

(a) 120/11 d (b) 140/13 d (c) 75/12 (d) 150/11 d (e) Not

A : B

E → 5 : 6

$W = t\times E = 5(x+8) : 6x$ ~~$5x+40+6x = 150$~~

$\frac{5(x+8)}{6x} = \frac{3}{2} \Rightarrow 5x + 40 = 9x$

$4x = 40$

$\boxed{x=10}$

T.W

$= 5x+40+6x = 50+40+60 = 150$ $= \left(\frac{150}{11}\right) \Rightarrow$ A&B T

(34) A and B together can do a piece of work in 60 days. A and C can do the same work in 45 days. The ratio of work efficiency of B and C is 1:2. In how many days they together can do the same work?

(a) 30 days (b) 24 days (c) 25 days (d) 36 days (e) NOT

	7	8
A+B → $\boxed{60 \text{ days}}$	4	3
A+C → $\boxed{45 \text{ days}}$	3	4

B : C = 1 : 2

A+B = 3

A + 1 = 3 ⇒ $\boxed{A=2}$

$\boxed{B=1}$ $\boxed{C=2}$

T.W = 60 × 3 = 180

A+B+C = 2+1+2 = 5

All together = $\frac{180}{5}$ = 36 days

(35) 40 men can complete a work in 48 days. If 64 men started working and after x days, 32 more men joined the work. If the remaining work is completed in $16\frac{2}{3}$ days. then find the value of x.

(a) 2 (b) 5 (c) 8 (d) 4 (e) 6

T.W

40×48 = 64×x + 96 × $\frac{50}{3}$

1920 = 64x + 1600 ⇒ 64x = 320

$\boxed{x=5}$

(36) A takes 24 day in completing a work alone Time taken by A in completing 1/3rd of the work is equal to the time taken by B in completing half of the work. How many days will be taken in completing the work if both A & B start working together?

(a) 21/3 days (b) 48 days (c) 40 days (d) 48/5 days

(e) NOT

A → 24 d

B → 16 d

48 units ⟨2, 3⟩ 5

Together = $\frac{48}{5}$ =

(37) A alone can do a work in 20 days. B is 25% more efficient than A. A and B started working and worked for 4 days, then C alone completed the remaining job in 22 days. In how many days C alone can complete the entire job?

(a) 30 (b) 35 (c) 40 (d) 42 (e) 45

A → 20 days Eff → 4

B → Eff → 5] 9

9×4 = 36

$\boxed{36 \text{ units}}$

20×4 = 80 T.W

80−36 = $\frac{44}{22}$ = 2 → Efficiency of C

C's working = $\frac{80}{2}$ = 40 days

(38) 'A' is 40% less efficient than 'B' who can do the same work in 20% less time than 'C'. If A and B together can complete 80% of work in 12 days, then in how many days 60% of the work can be completed by B and C together?

(a) 2 days (b) 4 days (c) 6 days (d) 8 days (e) 10 days

Time → 4 : 5

Efficiency → A : B : C

3 : 5 : 4 → 9

80% → 8×12

60% → $\frac{8\times12}{80} \times \frac{60}{9}$ = $\boxed{8 \text{ days}}$

(39) A, B and C can do a piece of ~~work~~ in 12 days, 16 days and 20 days. x

WORK AND WAGES

(39) A. B and C can do a piece of work in 12 days, 16 days and 20 days. respectively. After completing the work, they got Rs. 3525. If the wages are divided

the proportion of their work, then find the share of A?

(A) Rs. 1500 (b) 2100 (c) 1300 (d) NOT

A → 12 days, B → 16 days, C → 20 days → 240 units (T.W)

E: 20, 15, 12

W → A : B : C

Eff → $\underbrace{20x : 15x : 12x}_{47}$

$\frac{3525}{47} \times 20 = \boxed{1500}$

(40) The ratio of efficiency of A and B and C is 2:3:5. A and B will work on alternative days and C will work for all the days. If the work will be completed in 10 days and they get the total amount Rs 9000, then find the amount received by B?

(a) ₹ 1500 (b) ₹ 1800 (c) ₹ 2000 (d) ₹ 1200 (e) NOT

A : B : C

E → 2 : 3 : 5

T → ~~5~~ : ~~5~~ : 10

A+C, B+C: 2+5 = 7, 3+5 = 8, 15

W → 10 : 15 : 50

$\underbrace{2 : 3 : 10}_{15}$ → $\frac{9000}{15} = 600 \times 3 \to$ B's = 1800

days

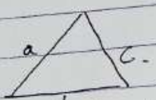
A's → 1, 3, 5, 7, 9

B's → 2, 4, 6, 8, 10

~~MEN~~ MENSURATION (2D & 3D)

2-Dimensional figures :→

Triangles

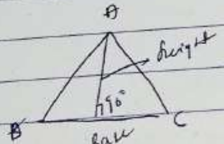


(Scalene triangle)

Perimeter = Sum of boundary
$= a+b+c$

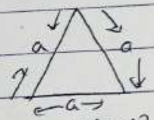
Area $= \sqrt{s(s-a)(s-b)(s-c)}$
(Hero's)

s = semiperimeter $= \frac{a+b+c}{2}$



Area $= \frac{1}{2} \times$ base $\times$ height

Equilateral Triangle

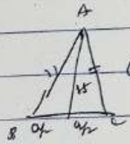


(Equilateral)

perimeter $= 3a$

Height $= \frac{\sqrt{3}}{2}a$

Area $= \frac{1}{2} \times b \times h = \frac{\sqrt{3}}{4}a^2$



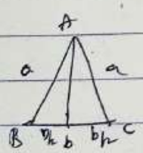
common

$H^2 + \left(\frac{a}{2}\right)^2 = a^2$

$H^2 = a^2 - \frac{a^2}{4} = \frac{3a^2}{4}$ $\boxed{H = \frac{\sqrt{3}\,a}{2}}$

Area $= \frac{1}{2} \times$ base $\times h$
$= \frac{1}{2} \times a \times \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{4}a^2$

Isosceles Triangle



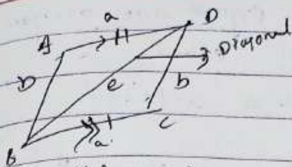
$A = \frac{1}{2} \times b \times h$

$A = \sqrt{s(s-a)^2(s-b)}$

$H = \sqrt{a^2 - \frac{b^2}{4}}$

$A = \frac{1}{2} \times b \times h = \frac{1}{2} \times b \times \sqrt{a^2 - \frac{b^2}{4}}$
$= \frac{b}{2}\sqrt{a^2 - \frac{b^2}{4}}$

Quadrilateral

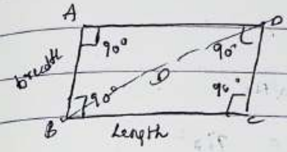


(parallelogram)

One pair of opposite sides are equal and parallel is called parallelogram

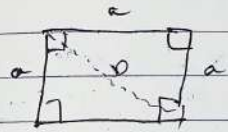
perimeter $= 2(a+b)$
A = base × height

Rectangle :→ parallelogram → Angles 90°



perimeter $= 2(l+b)$
Area = length × breadth
Diagonal = $D^2 = l^2 + b^2$
$D = \sqrt{l^2+b^2}$

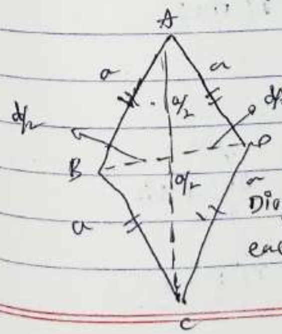
Square → Two pairs of opp are equal and 90°



→ Perimeter $= a+a+a+a = 4a$
Area $= a \times a = a^2$
Diagonal $= D^2 = \sqrt{a^2+a^2}$ $D^2 = \sqrt{2a^2}$
$D = \sqrt{2}a$

$D = \sqrt{2}a$
$D^2 = 2a^2 \Rightarrow \boxed{a^2 = \frac{D^2}{2}}$ → Area & Diagonal relationship

Rhombus :→ It square but not ~~having~~ 90°

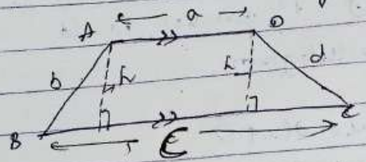


perimeter $= 4a$

$\boxed{A = \frac{1}{2} \times d_1 \times d_2}$

Trapezium :→

↳ only one pair of opposite sides are parallel equal.

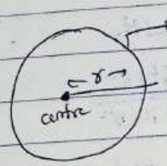


perimeter = a+b+c+d

Area = $\frac{1}{2}$ × height × (sum of ∥ sides)

$$\boxed{A = \frac{1}{2} \times \text{height} \times (a+c)}$$

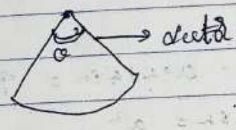
Circle :→



perimeter / circumference = $2\pi r$

Area = πr^2

Diameter = $2r$
↓
longest chord

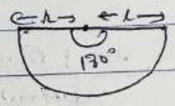


→ sector

Area = $\pi r^2 \times \frac{\theta}{360°}$ ← 90°

Circumference = $\frac{\theta}{360°} \times 2\pi r$

Semicircle



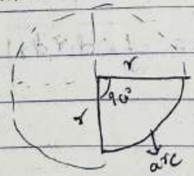
Area = $\frac{\theta}{360} \times \pi r^2 = \frac{180 \times \pi r^2}{360}$

$= \frac{\pi r^2}{2}$

Circumference = $\frac{2\pi r}{2} + 2r$

$= r(\pi+2)$

Quadrant



Area = $\frac{\pi r^2}{4}$

Circumference = $\frac{2\pi r}{4} + 2r$ ← arc

$= r\left(\frac{\pi}{2}+2\right)$

Questions of on 2-D

① The area of a square and a rectangle are equal. If the length of the rectangle is 8 more than the side of square and its breadth is 6m less than side of the square. What is the perimeter of rectangle?

ⓐ 100m ⓑ 96m ⓒ 110m ⓓ 80m ⓔ 120m

square   rectangle

$a^2 = l \times b$

$l = (a+8)$ $b = (a-6)$

$a^2 = (a+8)(a-6)$

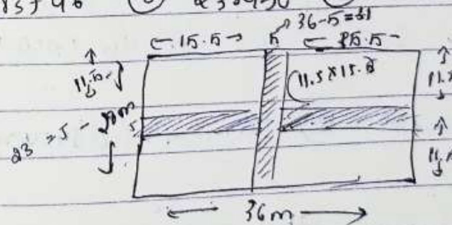
$a^2 = a^2 + 2a - 48$

$\Rightarrow 2a = 48$ $\boxed{a=24}$ $l = a+8 = 32$
$b = a-6 = 18$

$P = 2(l+b)$
$= 2 \times 50 = 100m$

② A 36m long and 27m broad rectangular plot has two concrete roads 5m wide running in the middle of the park, one parallel to the length and other parallel to the breadth. What would be the total cost of gravelling the plot, excluding the area covered by the roads @ 860 p. sq.m?

ⓐ 283740 ⓑ 233470 ⓒ 248860 ⓓ 265420 ⓔ 256680



→ Total = 4 (11.5 × 15.5)

Cross road = $(l+b-w)w$
$= (36+27-5)5 = 295$

Total area = $36 \times 27 = (32+4)(32-4)$
$= 32^2 - 4^2 = 1008$